

Algebraic Groups 1

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These notes are partially inspired by the talk *Algebraic Groups and Torsors* in AGGITatE 2026, and will separately cover the theory of algebraic groups over $\mathbb{C}$ and over local fields. The part on algebraic groups over $\mathbb{C}$ is a continuation of the Lie algebra notes, and the main reference is *Linear Algebraic Groups* by T.A.Springer. For algebraic groups over local fields, we will use *Algebraic Groups and Number Theory* by Vladimir Platonov & Andrei Rapinchuk. Further we will study arithmetic groups and see the applications in number theory.

1 Basic Properties

In this section we present some basic results on algebraic groups over algebraically closed fields as well as over arbitrary fields, which will be used constantly throughout the notes

Definition 1.1. An **algebraic group** is an algebraic variety G that is also a group scheme. If G and the morphisms from the structure of group schemes are defined over k and the identity element e is a k -rational point, we say G is a k -group.

G is called a **linear algebraic group** if it is affine, or equivalently, it is a closed subgroup of $GL_n(k)$ for some n

We will not elaborate on group schemes and its functor point of view, but it is certainly useful in some cases. An accessible source is [Wat79], and we will follow the notations there.

Example 1. $G = \mathbb{G}_a = \mathbb{A}_k^1$, the **additive group**. As a functor, it is represented by $k[t]$. The comultiplication is given by $\Delta : k[t] \rightarrow k[u] \otimes k[v] \cong k[u, v]; t \mapsto u + v$, the antipode $\iota : k[t] \rightarrow k[t]$ sends $t \mapsto -t$, and the counit $e : k[t] \rightarrow k$ sends t to 0

Example 2. $G = \mathbb{G}_m = \mathbb{A}_k^{1*}$, the **multiplicative group**. As a functor, it is represented by $k[t^{\pm 1}]$. The comultiplication is given by $\Delta : k[t^{\pm 1}] \rightarrow k[u^{\pm 1}] \otimes k[v^{\pm 1}] \cong k[u^{\pm 1}, v^{\pm 1}]; t \mapsto uv$, the antipode $\iota : k[t^{\pm 1}] \rightarrow k[t^{\pm 1}]$ sends $t \mapsto t^{-1}$, and the counit $e : k[t^{\pm 1}] \rightarrow k$ sends t to 1

If $g \in G$, then the maps $x \mapsto gx$ and $x \mapsto xg$ are isomorphisms of the variety G . We shall frequently use this observation.

Proposition 1.1. 1. There is a unique irreducible component G^0 of G that contains the identity element e . It is a closed normal subgroup of finite index;

2. G^0 is the unique connected component of G containing e ;

3. Any closed subgroup of G of finite index contains G^0 .

Proof. (1) Let X, Y be two irreducible components containing e and $\mu : G \times G \rightarrow G$. Then $XY = \mu(X \times Y)$ is irreducible and so is its closure $\overline{XY}$ (and so an irreducible component). Since X, Y are contained in $\overline{XY}$, it follows that $X = Y = \overline{XY}$. Hence X is closed under multiplication.

Now the taking inverse map $i : x \mapsto x^{-1}$ is a homeomorphism, and so X^{-1} is an irreducible component containing e . We deduce that $X^{-1} = X$ and X by preceding argument and that X is a closed subgroup. Similarly, using that the inner automorphisms are homeomorphisms, we get $xXx^{-1} = X$ for $x \in G$ and so X is a normal subgroup. Note that xX are irreducible components of G , and then noetherianity implies that X has finite index

(2) We see that the irreducible components (cosets of G^0) of G are mutually disjoint. It then follows that the irreducible components must coincide with the connected components

(3) If H is a closed subgroup of G of finite index, then H^0 is a closed subgroup of G^0 of finite index. Hence H^0 is both close and open in G^0 , and so $H^0 = G^0$ since G^0 is connected by (2) $\square$

Recall that for a variety over algebraically closed field, the smooth locus is a nonempty open subset. For $g_1, g_2 \in G$, the map $x \mapsto g_2 g_1^{-1} x$ is a morphism of varieties sending $g_1 \mapsto g_2$. We deduce that

Proposition 1.2. *Algebraic groups are smooth varieties.*

Now for smooth varieties, irreducible and connected components always coincide, and this gives an alternative proof of proposition 1.1. We always denote G^0 the component of the algebraic group G containing the identity, which is therefore a open and closed normal subgroup of G of finite index, and so $\dim G = \dim G^0$

Definition 1.2. *The tangent space $T_e(G)$ at e is called the **Lie algebra** $L(G)$ of G*

By the definition of smoothness, $\dim L(G) = \dim G$. If G is a linear algebraic group, i.e. $G \subset GL_n(k)$ for some n , then $L(G) \subset M_n(k) = L(GL_n(k))$ (see example 2 in the Lie algebra notes), and the Lie bracket is given by the standard formula $[x, y] = xy - yx$. For any $g \in G$, we have $gL(G)g^{-1} = L(G)$, giving rise to a morphism of algebraic groups $G \rightarrow GL(L(G))$ by $g \mapsto \varphi_g$ where φ_g send x to gxg^{-1} for $x \in L(G)$. This is the **adjoint representation of G** and is written Ad .

Recall we also have $ad : L(G) \rightarrow \text{End}(L(G))$ by $ad_x : y \mapsto [x, y]$ for $x \in L(G)$, called the **adjoint representation of the Lie algebra $L(G)$** . In fact, ad is the differential of the representation Ad at e . We also have the **Killing form**, a symmetric bilinear form on $L(G)$, given by $(x, y) = \text{tr}(ad_x \cdot ad_y)$. Note that the Killing form is **invariant** under the adjoint action of G , i.e. $(x, y) = (\varphi_g x, \varphi_g y)$ for $g \in G$. Differentiate and we get $([x, y], z) = (x, [y, z])$, i.e. the Killing form is an invariant symmetric bilinear form on $L(G)$.

1.1 Jordan Decomposition

Recall a matrix g is **semisimple** if it is diagonalizable; g is **unipotent** if $g - I$ is nilpotent, i.e. all the eigenvalues of g are 1

Proposition 1.3. *(Additive Jordan Decomposition) Let $g \in M_n$. There exist unique $g_s, g_n \in M_n$ with g_s semisimple and g_n nilpotent, such that $g = g_s + g_n$ and $g_s g_n = g_n g_s$*

Proof. See [T.A10, proposition 2.4.4] □

Corollary 1.1. *(Multiplicative Jordan Decomposition) Let $g \in GL_n$. There is $g_s, g_u \in GL_n$ with g_s semisimple and g_u unipotent such that $g = g_s g_u = g_u g_s$*

Proof. Write $g = g_s g_n$ as proposition 1.3. In the proof of proposition 1.3 we have that the eigenspaces of g_s are precisely the generalized eigenspaces of g , therefore $g_s \in GL_n$. Hence we can take $g_u = 1 + g_s^{-1} g_n$. The uniqueness follows from the uniqueness of g_s and g_n □

Now we lay out some results on the unipotent groups (i.e. linear algebraic groups whose elements are unipotent). Recall that U_n is the group of unipotent upper triangular matrices, i.e., the subgroup of $\mathfrak{b}$ whose elements have diagonal entries 1

Proposition 1.4. *Let $G \subset GL_n$ be a unipotent group. Then there is $x \in GL_n$ such that $xGx^{-1} \subset U_n$*

Proof. We use Burnside's theorem which states that the image of a irreducible representation spans the entire endomorphism ring. Put $V = k^n$. Then G is a group of unipotent linear maps of V . If V has a subrepresentation, we can apply induction on n . Otherwise, V is an irreducible representation of G . By Burnside's theorem, elements in G span $\text{End}(V)$.

Take any $g \in G$. Then $\text{tr}(g) = n$ and $\text{tr}((1 - g)h) = 0$ for any $h \in G$, and hence for any $h \in \text{End}(V) = M_n$. We deduce that $1 - g = 0$ and so $G = \{1\}$ and the assertion is trivial. □

Corollary 1.2. *Unipotent groups are nilpotent, and hence solvable.*

Proof. This is because U_n is nilpotent. □

Consequently, if G is a unipotent group and $G \rightarrow GL(V)$ is a rational representation, there is a vector $0 \neq v \in V$ fixed by all of G

1.2 G -spaces

A G -space is a variety X on which G acts as a permutation group, the action being given by a morphism of varieties

Lemma 1.1. 1. *An orbit $G \cdot x$ is open in its closure*

2. *There is a closed orbit*

Proof. See [T.A10, lemma 2.3.3] □

Let X be an affine G -space with action $a : G \times X \rightarrow X$. We have $k[G \times X] = k[G] \otimes k[X]$. The pullback through $a^* : k[X] \rightarrow k[G] \otimes k[X]$ is then an algebra homomorphism. For $g \in G, x \in X, f \in k[X]$, define

$$s(g) : k[X] \rightarrow k[X] \text{ by } s(g)f(x) = f(g^{-1}x)$$

Then $s(g)$ is an invertible linear map of the (in general infinite dimensional) vector space $k[X]$ and $s : G \rightarrow GL(k[X])$ is a representation of G . The next result will imply that s can be built up from rational representations

Proposition 1.5. *Let V be a f.d. subspace of $k[X]$*

1. *There is a f.d. subrepresentation of $k[X]$ containing V*

2. *V is a subrepresentation iff $a^*V \subset k[G] \otimes V$. If this is so, s defines a map $s_V : G \times V \rightarrow V$ which is a rational representation of G*

Proof. For (1), we may assume that $V = kf$ a dim-1 space for some $f \in k[X]$. Let $a^*f = \sum u_i \otimes f_i$ for some $u_i \in k[G], f_i \in k[X]$. Then $s(g)f(x) = f(g^{-1}x) = a^*f(g^{-1}, x) = \sum u_i(g^{-1})f_i(x)$ and we see that $s(g)f$ is in the span of f_i (which is f.d.). Then the subspace W spanned by $s(g)f$ for $g \in G$ has the properties of (1)

By a similar argument we see that if $a^*V \subset k[G] \otimes V$ then V is $s(G)$ -stable. Conversely, assume V is $s(G)$ -stable. Let f_i be a basis for V and extend it to a basis f_i, g_j for $k[X]$. For $f \in V$, write

$$a^*f = \sum u_i \otimes f_i + \sum v_j \otimes g_j$$

where $u_i, v_j \in k[G]$. Then, as above,

$$s(g)f = \sum u_i(g^{-1})f_i + \sum v_j(g^{-1})g_j$$

By assumption, we deduce that $v_j(g^{-1}) = 0$ for all g , and hence v_j vanishes. This proves (2). □

By considering G acts by left and right translation on itself, we are able to show that linear algebraic groups are indeed closed subgroups of some GL_n . See [T.A10, theorem 2.3.7].

Let V be a not necessarily finite dimensional vector space over k . We say $a \in \text{End}(V)$ is **locally finite** if V is a union of f.d. a -stable subspaces. Then proposition 1.5 (1) essentially shows that G acts locally finitely on a G -space X

We enclose this section with a theorem by Kostant -Rosenlicht:

Proposition 1.6. *Let G be a unipotent group and X an affine G -space. Then all orbits of G in X are closed.*

Proof. Let O be an orbit. Replacing X with $\overline{O}$ if necessary, we may assume that O is dense in X . By lemma 1.1, O is open. Let Y be its complement and $\mathcal{I}_X(Y)$ be the defining ideal. Then G acts locally finitely on $\mathcal{I}_X(Y)$ and by the observation after corollary 1.2, there is a non-zero function $f \in \mathcal{I}_X(Y)$ fixed by all of G . Hence f is constant on the orbit O . Since O is dense in X , f is constant on X (i.e. a constant), and so $\mathcal{I}_X(Y) = k[X]$ and $O = X$, as required $\square$

2 Structure of Commutative Algebraic Groups

Theorem 2.1. *(Jordan Decomposition) Let G be a commutative linear algebraic group. The sets G_s, G_u of semisimple and unipotent elements of G are closed subgroups, and G is isomorphic to their direct product. Moreover, if G is connected, so is G_s, G_u*

Proof. We may assume G is a closed subgroup of some GL_n . Clearly G_s, G_u are groups. By [T.A10, lemma 2.4.2], we can find a basis so that G lies in $\mathfrak{n}$ and $G_s = G \cap D_n$, where D_n is the set of diagonal matrices, and so G_s is closed in G . Likewise, $G_u = G \cap U_n$ is closed.

The uniqueness of Jordan decomposition shows that the product map $\pi : G_s \times G_u \rightarrow G$ is an isomorphism of abstract groups. It is also a morphism of algebraic varieties, since the map $g \mapsto g_s$ essentially picks out the diagonal entries of g . By considering the projection maps, this proves the corollary. $\square$

Proposition 2.1. *Let G be a connected linear algebraic group of dimension 1. Then G is commutative, and $G = G_u$ or $G = G_s$. Moreover, if $G = G_u$ is unipotent, and the base field k has character $p > 0$, then the elements in G have order dividing p .*

Proof. Fix $g \in G$ and consider the morphism $\phi : x \mapsto gxg^{-1}$. Then the closure $\overline{\phi G}$ is an irreducible closed subset of G . Hence it is either the singleton $\{e\}$ or all of G . If it is all of G , then the complement $G - \phi G$ is finite. Let G be the closed subgroup of some GL_n . Then there are only finite possibilities for the characteristic polynomials $\det(T \cdot I - x)$ as x runs over G . The connectedness of G implies that the characteristic polynomial is constant, and equal to $(T - 1)^n$. This means that G is unipotent and hence solvable. The commutator subgroup $G_1 = [G, G]$ is a connected and closed subgroup, and hence $G_1 = \{e\}$ for dimension reason and G is commutative. The product $G = G_s \times G_u$ forces $G = G_s$ or $G = G_u$. Finally, if $G = G_u$ and $\text{char}(k) = p > 0$, G^{p^k} generated by the p^k -th power of elements of G is a connected closed subgroup, and, again, can only be $\{e\}$ or G . Since G is a subgroup of GL_n , $G^{p^h} = \{e\}$ for $p^h \geq n$. This forces $G^p = \{e\}$. $\square$

2.1 Diagonalizable Groups and Tori

Definition 2.1. A **torus** is an algebraic group isomorphic to $D_n = \mathbb{G}_m^n$ for some n . A **vector group** is an algebraic group isomorphic to $\mathbb{G}_a^n$ for some n .

Definition 2.2. A **diagonalizable group** is one isomorphic to a closed subgroup of some torus.

Before discussing diagonalizable groups in general, we will study tori in more detail.

Example 3. We have seen in the Toric Variety course that a **(rational) character** of G is a homomorphism $\chi : G \rightarrow \mathbb{G}_m$. We write $X^*(G) = \text{Hom}_{\text{alg}}(G, \mathbb{G}_m)$, the set of characters. If G is commutative then $X^*(G)$ also has a structure of abelian group.

A **cocharacter** is a homomorphism $\mathbb{G}_m \rightarrow G$, and we write $X_*(G) = \text{Hom}_{\text{algrp}}(\mathbb{G}_m, G)$ the set of cocharacters.

Let $G = D_n$ a torus. For $x \in D_n$, write it as $\text{diag}(\chi_1(x), \dots, \chi_n(x))$. Then χ_i are characters of D_n . Then $k[D_n] = k[\chi_1^\pm, \dots, \chi_n^\pm]$. Dedekind's theorem tells that any character comes from a monomial $\chi_1^{a_1} \dots \chi_n^{a_n}$ for $(a_1, \dots, a_n) \in \mathbb{Z}^n$. Hence $X^*(D_n) \cong \mathbb{Z}^n$, called the character lattice. Similarly, we can show that a homomorphism $\mathbb{G}_m \rightarrow D_n$ is of form $x \mapsto \text{diag}(x^{a_1}, \dots, x^{a_n})$ for $(a_1, \dots, a_n) \in \mathbb{Z}^n$. Hence $X_*(D_n) \cong \mathbb{Z}^n$, called the cocharacter lattice.

Example 4. By contrast, a vector group G has no non-trivial characters. The homomorphisms from G into $\mathbb{G}_a$, called additive functions, are the same as they are in linear algebra, that is, they are just k -vector space maps from $G \cong k^n$ to k . Monomials in these maps (but not just the maps themselves) furnish a k -basis of $k[G]$.

Theorem 2.2. *TFAE for a linear algebraic group G :*

1. G is diagonalizable
2. $X^*(G)$ is a f.g. abelian group that spans $k[G]$
3. Any rational representation of G is a direct sum of dim-1 representations.

Proof. (1) $\implies$ (2): if G is a closed subgroup of some D_n , then $k[G]$ is a quotient of $k[D_n]$. Since the restriction of a character of D_n is a character of G , we see that the restrictions of characters of D_n span $k[G]$. By Dedekind's theorem they form a basis and any character of G is a linear combination of these restrictions. Hence the restriction map $X^*(D_n) \rightarrow X^*(G)$ is surjective, and $X^*(G)$ is f.g. by the images of χ_i

(2) $\implies$ (3): Let $\phi : G \rightarrow GL(V)$ be a rational representation. Since $X^*(G)$ spans $k[G]$, we can find suitable linear maps A_χ of V for χ characters such that

$$\phi(x) = \sum_{\chi \in X^*(G)} A_\chi \chi(x)$$

for $x \in G$. The relation $\phi(xy) = \phi(x)\phi(y)$ together with Dedekind's theorem implies that $A_\chi A_\varphi = \delta_{\chi\varphi} A_\chi$ for χ, φ characters. By letting $x = 1$, we also get $\sum A_\chi = 1$. Put V_φ the image of A_φ . Then

$$\phi(x)V_\varphi = \sum_{\chi} \chi(x) A_\chi A_\varphi V = \chi(x)V_\varphi$$

by the properties of A_χ . We deduce that V is the direct sum of V_χ and X acts on V_χ by multiplication by the scalar $\chi(x)$. This proves (3).

(3) $\implies$ (1) is immediate after realizing G a closed subgroup of some $GL(V)$ □

Now let G be a diagonalizable group. As $X^*(G)$ is f.g., we can write $X^*(G) \cong \mathbb{Z}^n \times M$ where M is a finite abelian group. Write $A = X^*(G)$. Then one easily checks that the group algebra $k[A] \cong k[X_1^\pm, \dots, X_n^\pm] \otimes k[M] \cong k[D_n \times M]$. Since $k[G]$ is spanned over k by $X^*(G)$, we have $k[G] \cong k[A]$, whence G is isomorphic to $D_n \times M$. We conclude that

Corollary 2.1. *Any diagonalizable group G is the direct product of a torus and a finite abelian group, with the order of the latter prime to the characteristic of k if this is positive. G is a torus if and only if it is connected, or if and only if $X^*(G)$ is free abelian.*

This is immediate using that k has no nontrivial p -th root of unity when $\text{char}(k) = p$

Proposition 2.2. (*Rigidity of diagonalizable groups*) Let G, H be diagonalizable groups and let V be a connected variety. Assume we are given a morphism of varieties $\phi : G \times V \rightarrow H$ such that for any $v \in V$ the map $x \mapsto \phi(x, v)$ is a homomorphism from G to H . Then $\phi(x, v)$ is constant in V .

Proof. For $\varphi \in X^*[H]$, $\varphi(\phi(x, v))$ can be written in the form

$$\varphi(\phi(x, v)) = \sum_{\chi \in X^*(G)} f_{\chi, \varphi}(v) \chi(x)$$

where $f_{\chi, \varphi} \in k[V]$. For a fixed v the rhs is then a character of G . Dedekind's theorem shows that $f_{\chi, \varphi}(v) = 1$ for one χ and 0 for the others, whence $f_{\chi, \varphi}^2 = f_{\chi, \varphi}$. The connectedness of V forces $f_{\chi, \varphi} = 1$ for one χ and 0 for the others $\square$

We give an application of the proposition. If G is an arbitrary linear algebraic group and H is a closed subgroup, let $Z_G(H)$ and $N_G(H)$ the centralizer and normalizer of H in G . They are closed subgroups of G and $Z_G(H)$ is normal in $N_G(H)$

Corollary 2.2. If H is diagonalizable, then $Z_G(H)$ has finite index in $N_G(H)$ and their identity components coincide, i.e. $N_G(H)^0 = Z_G(H)^0$

Proof. Apply the proposition by letting $V = N_G(H)^0$, ϕ the homomorphism sending $(x, y) \mapsto yxy^{-1}$ from $H \times V$ to H , and we see that elements in $N_G(H)$ are central. Now use proposition 1.1 (1). $\square$

2.2 Unipotent Groups

We have seen that a commutative group consisting of semisimple elements is diagonalizable and thus the direct product of a torus and a finite abelian group. We now give a similar structural theorem for unipotent groups.

Definition 2.3. We say a unipotent group is **elementary** if it is commutative, and its elements are of order dividing p when $\text{char}(k) = p$

Theorem 2.3. An elementary unipotent group is a vector group $\mathbb{G}_a^n$ when $\text{char}(k) = 0$, and is the direct product of a vector group and a product of copies of $\mathbb{Z}/p$ when p is positive. In particular, G is a vector group if and only if it is connected.

Proof. See [T.A10, theorem 3.4.7] $\square$

Corollary 2.3. Let G be a connected linear algebraic group of dimension one. Then G is isomorphic to either $\mathbb{G}_a$ or $\mathbb{G}_m$.

Proof. Proposition 2.1 tells that G is commutative. Therefore we can directly apply corollary 2.1 or theorem 2.3 depending on G is diagonalizable (elements are semisimple) or unipotent. $\square$

3 Derivations, Deifferentials and Lie algebras

We will give a quick recap on the notions in geometry and Lie algebras in this section.

For an affine variety X , the **tangent space at** $x \in X$ is $T_x X = \text{Hom}(m_x/m_x^2, k) \cong \text{Der}_k(k[X], k)$, where m_x is the kernel of $k[X] \rightarrow k; f \mapsto f(x)$, i.e. functions vanishing at x , and k is the $k[X]$ -module via $f \mapsto f(x)$. Let $\phi : X \rightarrow Y$ be a morphism of affine varieties with corresponding homomorphism $\phi^* : k[Y] \rightarrow k[X]$. It further induces a linear map of tangent spaces

$$d\phi_x : T_x X = \text{Der}_k(k[X], k) \rightarrow T_{\phi(x)} Y = \text{Der}_k(k[Y], k)$$

This is the **differential of** ϕ at x , or the tangent map at x . If ϕ is an isomorphism, then so is $d\phi_x$

Definition 3.1. We say $x \in X$ is **simple** or **nonsingular**, or X is smooth at x if $\dim T_x X = \dim X$.

Given a commutative ring R and an R -module A , consider the multiplication map $m : A \otimes_R A \rightarrow A$ given by $a \otimes b \mapsto ab$. Then $I = \ker(m)$ is generated by $a \otimes 1 - 1 \otimes a$ for $a \in A$. The **module of differentials** of R -algebra A is defined by $\Omega_{A/R} = I/I^2$. One checks that we have an R -derivation $d : A \rightarrow \Omega_{A/R}$ via $d(a) = a \otimes 1 - 1 \otimes a$ which make $\Omega_{A/R}$ an A -module.

Theorem 3.1. For any A -module M , every R -derivation $\phi : A \rightarrow M$ uniquely factors through d , i.e. there is a unique A -module homomorphism $\varphi : \Omega_{A/R} \rightarrow M$ such that $\phi = \varphi \circ d$, i.e. $\text{Hom}_A(\Omega_{A/R}, M) \cong \text{Der}_R(A, M)$

The theorem shows that $(\Omega_{A/R}, d)$ is the unique universal module for R -derivations of A .

Let A be a f.g. R -algebra of form $A = R[T_1, \dots, T_m]/(f_1, \dots, f_n)$. Let $t_i = \overline{T_i} \in A$ and write $t = (t_1, \dots, t_m)$. Denote partial differentiation with respect to T_i by D_i . Then a straightforward calculation essentially yields

Lemma 3.1. dt_i generate $\Omega_{A/R}$. The kernel of the homomorphism $\phi : A^m \rightarrow \Omega_{A/R}$ via $\phi e_i = dt_i$ ($\{e_i\}$ is the standard basis for A^m) is the submodule generated by $\sum D_i(f_j(t))dt_i$ for $1 \leq j \leq n$

In particular, if $A = R[T_1, \dots, T_m]$ is a polynomial ring, then $\Omega_{A/R} \cong A^m$ is a free A -module generated by dT_i

Next we discuss the case of fields. Let F be a field and E, E' two finite extensions of F with $E' \subset E$. Note that we have an exact sequence of derivations by restricting:

$$0 \rightarrow \text{Der}_{E'}(E, E) \rightarrow \text{Der}_F(E, E) \rightarrow \text{Der}_F(E', E)$$

By theorem 3.1, we get

$$0 \rightarrow \text{Hom}_E(\Omega_{E/E'}, E) \rightarrow \text{Hom}_E(\Omega_{E/F}, E) \rightarrow \text{Hom}_{E'}(\Omega_{E'/F}, E)$$

Also note that there is a natural isomorphism $\text{Hom}_{E'}(\Omega_{E'/F}, E) \cong \text{Hom}_E(E \otimes_{E'} \Omega_{E'/F}, E)$. Plug it in and taking the dual, we get an exact sequence of differentials

$$E \otimes_{E'} \Omega_{E'/F} \rightarrow \Omega_{E/F} \rightarrow \Omega_{E/E'} \rightarrow 0$$

This is the **canonical exact sequence of modules of differentials**

Lemma 3.2. The sequence is left exact if E is separable over E'

Proof. Dualizing, we see that it suffices to show that the restriction $\text{Der}_F(E, E) \rightarrow \text{Der}_F(E', E)$ is surjective, i.e. every F -derivation of E' into E extends to E . Assume E/E' is a simple extension and $E \cong E(\alpha) \cong E'[x]/f$ for some irr. polynomial f with $f(\alpha) = 0, f'(\alpha) \neq 0$. Let $D \in \text{Der}_F(E', E)$. For $g \in E$, write $g = \sum a_i x^i$ for some $a_i \in E'$ and define $Dg = \sum Da_i \cdot x^i$. Then D extends to a F -derivation D' of E into E . Note that $D'\alpha = a$ iff $0 = D(f(\alpha)) = (Df)(\alpha) + f'(\alpha)a$, which has a solution for a since $f'(\alpha) \neq 0$. This proves the lemma. $\square$

Corollary 3.1. If $E = F(x)$, we have $\dim_E \Omega_{E/F} \leq 1$. $\Omega_{E/F} = 0$ iff E is algebraic separable over F

Remark 3.1. This is unramification in geometry

In proposition 1.2 we roughly showed that algebraic groups are smooth using the existence of smooth points on varieties and that translations are isomorphisms. In fact, most points on varieties are smooth

Theorem 3.2. Let X be a irreducible varieties of dimension e .

1. For x a simple point on X , there is an affine neighborhood U of x such that $\Omega_U = \Omega_{k[U]/k}$ is a free $k[U]$ -module with basis dg_i for some $g_i \in k[U]$
2. The simple points of X form a non-empty open set
3. For $x \in X$, $\dim T_x X \geq e$

Proof. See [T.A10, theorem 4.3.3] □

Definition 3.2. A morphism $\phi : X \rightarrow Y$ is **dominant** if ϕX is dense in Y . If so, the coordinate ring $k[Y]$ embeds into $k[X]$, hence the function fields $k(Y)$ embeds into $k(X)$. We say ϕ is **separable** if the field extension $k(X)/k(Y)$ is separable in this case.

Theorem 3.3. Let $\phi : X \rightarrow Y$ be a morphism of irreducible varieties

1. Assume that x is a simple point of X such that $\phi(x)$ is a simple point of Y and that $d\phi_x$ is surjective. Then ϕ is dominant and separable
2. If ϕ is dominant and separable, then the points of X satisfying property of (1) form a non-empty open subset of X .

Proof. By theorem 3.2, we may assume that X and Y are both affine smooth varieties, whence $\Omega_X = \Omega_{k[X]/k}$ and Ω_Y are free $k[X]$ -module and $k[Y]$ -module with rank $\dim X = d$ and $\dim Y = e$ respectively. The universal property of module of differentials gives that $\phi^* : k[Y] \rightarrow k[X]$ induces a $k[Y]$ -module homomorphism $(\phi^*)^0 : \Omega_Y \rightarrow \Omega_X$, and therefore a $k[X]$ -module homomorphism $\varphi : k[X] \otimes_{k[Y]} \Omega_Y \rightarrow \Omega_X$. Fix basis for these two free $k[X]$ -modules, and φ can be described by a $d \times e$ matrix A . By assumption take $x \in X$ with $\phi(x)$ a simple point and $d\phi_x$ surjective. Then A evaluating at x has rank e , whence the rank of A is at least e . Since the rank is at most e it must equal e . It follows that φ is injective, and so is $(\phi^*)^0$. As Ω_X, Ω_Y are free modules, ϕ^* is injective and so ϕ is dominant. Also letting $E = k[X], E' = k[Y], F = k$ in lemma 3.2, we see that ϕ is separable.

Conversely, if ϕ is separable and dominant, then the rank of A (entries in $k(X)$) equals e . The set of x such that A evaluating at x has rank e is therefore open and nonempty. □

Recall that a **homogeneous space** for G is a G -space on which G acts transitively.

Theorem 3.4. Let G be a connected algebraic group

1. Any homogeneous X for G is irreducible and smooth
2. Let $\phi : X \rightarrow Y$ be a G -morphism of homogeneous spaces. Then ϕ is separable iff $d\phi_x$ is surjective for some $x \in X$ iff $d\phi_x$ is surjective for all $x \in X$
3. Let $\phi : G \rightarrow G'$ a homomorphism of algebraic groups. Then ϕ is separable iff $d\phi_e$ is surjective.

Proof. If X is as in (1), for $x \in X$ the morphism $G \rightarrow X; g \mapsto g \cdot x$ is then surjective and so X is irreducible. Also for any $g \in G$ the endomorphism of X $x \mapsto g \cdot x$ is then an isomorphism, hence $x \in X$ simple implies $g \cdot x$ simple. By theorem 3.2 (2), we get that X is smooth. Now (2) follows from theorem 3.2 since ϕ is surjective, and (3) follows from (2) by regarding G and G' as homogeneous spaces of G . □

Recall that the tangent space $T_e G = L(G)$ is the Lie algebra of an algebraic group G . Now we show that the module $\Omega_G = \Omega_{k[G]/k}$ is controlled by $L(G)$. Denote by λ and ρ the representation of left and right translations of G . Let $A = k[G]$ and $m : A \otimes A \rightarrow A$ the multiplication. For $F \in k[G \times G] = A \otimes A$,

$m \circ F(x) = F(x, x)$. So $I = \ker(m)$ is the ideal of functions in $k[G \times G]$ vanishing on the diagonal. It is now clear that for $x \in G$ $\lambda(x) \otimes \lambda(x)$ and $\rho(x) \otimes \rho(x)$ are automorphisms of $k[G \times G]$ and stabilize I, I^2 , and so are also automorphisms of $\Omega_G = I/I^2$. We still denote them by $\lambda(x), \rho(x)$

Also recall that the automorphism ad_x of $x \in G$ is an automorphism of $T_e G$ induced by the automorphism $\varphi_x : y \mapsto xyx^{-1}$ of G . Let M_e be the functions of $k[G]$ vanishing at e . Then $(T_e G)^* \cong M_e/M_e^2$. For $f \in k[G]$, let $\delta f = f - f(e) + M_e^2 \in (T_e G)^*$. We then have $(\delta f)X = X(f)$ for $X \in T_e G = \text{Der}_k(A, k)$

Proposition 3.1. *There is an isomorphism of $k[G]$ -module $\phi : \Omega_G \xrightarrow{\sim} k[G] \otimes_k (T_e G)^*$ such that $\phi \circ \lambda(x) \circ \phi^{-1} = \lambda(x) \otimes 1$ and $\phi \circ \lambda(x) \circ \rho^{-1} = \rho(x) \otimes (ad_x)^*$. If $f \in k[G]$ and $\Delta f = \sum f_i \otimes g_i$, then $\phi(df) = -\sum f_i \otimes \delta g_i$*

Proof. Consider the map sending $(x, y) \mapsto (x, xy)$ on G . It is an automorphism of $G \times G$ and induces an automorphism φ of $k[G \times G]$ such that $\varphi F(x, y) = F(x, xy)$ for $F \in k[G \times G]$. Hence φI is the ideal of function vanishing on $G \times \{e\}$, which is precisely $k[G] \otimes M_e$. Then $\varphi I^2 = k[G] \otimes M_e^2$, and so φ induces an isomorphism of $I/I^2 = \Omega_G$ and $k[G] \otimes M_e/M_e^2 \cong k[G] \otimes_k (T_e G)^*$. The rest is direct verification. $\square$

Now let $\mathcal{D} = \text{Der}_k(A, A)$. Then $\mathcal{D}$ has the Lie algebra structure through the commutator bracket. Then λ and ρ define a representation of G in $\mathcal{D}$ via

$$\lambda(x)D = \lambda(x) \circ D \circ \lambda(x)$$

for $x \in G, D \in \mathcal{D}$. Consider the linear map $\alpha = \alpha_G : \mathcal{D} \rightarrow T_e G$ with $(\alpha_G D)(f) = (Df)(e)$. Next proposition gives the Lie algebra structure on $T_e G$

Proposition 3.2. *α induces an isomorphism between $T_e G$ and the set of $D \in \mathcal{D}$ commuting with all $\lambda(x)$ for $x \in G$. We have that for $x \in G$, $\alpha \circ \rho(x) \circ \alpha^{-1} = ad_x$.*

Proof. See [T.A10, proposition 4.4.5] $\square$

Example 5. (1) Let $G = \mathbb{G}_a$. Then $A = k[T]$. The derivations of $k[T]$ commuting with translations $T \mapsto T + a$ are just multiples of $X = d/dT$. So $\mathfrak{g}$ is just the dim-1 Lie algebra kX with $[X, X] = 0$

(2) Let $G = \mathbb{G}_m$. Then $A = k[T^{\pm 1}]$. The derivations of $k[T^{\pm 1}]$ commuting with translations $T \mapsto aT$ are just multiples of $X = T \cdot d/dT$. Again, $\mathfrak{g}$ is just a dim-1 Lie algebra

(3) Let $G = GL_n$. Then $A = k[T_{ij}, \det(T_{ij})^{-1}]$. Denote $\mathfrak{gl}_n$ the Lie algebra of $n \times n$ matrices with commutator bracket. For $X = (x_{ij}) \in \mathfrak{gl}_n$, define a derivation D_X by

$$D_X T_{ij} = -\sum_{h=1}^n T_{ih} x_{hj}$$

D_X commutes with all left translations and so lies in the Lie algebra of GL_n . Since $X \mapsto D_X$ is injective, $L(G)$ consists of D_X since $\dim L(G) = \dim GL_n = \dim \mathfrak{gl}_n = n^2$. Therefore we can identify $L(G)$ with $\mathfrak{gl}_n$.

More examples of algebraic groups and their Lie algebras can be found in the Lie Algebra notes.

4 Some Algebraic Geometry

Now we follow chapter 5 of [T.A10] to give some results of algebraic geometry and their applications in algebraic groups. One main item is the construction of the quotients of linear algebraic groups as closed subgroups.

Across this section, X, Y are two irreducible varieties over an algebraically closed field k . Recall that a morphism $\phi : X \rightarrow Y$ is dominant iff the pullback $\phi^* : k[Y] \rightarrow k[X]$ is injective, and so induces a function field extension. We say ϕ is **birational** if $k[X] = k[Y]$. The next lemma gives an equivalent description of birational morphisms that one should be familiar with.

Lemma 4.1. *$\phi : X \rightarrow Y$ is birational iff there is a nonempty open subset U of X such that $\phi|_U$ is open and ϕ induces an isomorphism of varieties $U \cong \phi U$*

Proof. Assume ϕ is birational and X, Y are affine. Then $k[X] = k[Y][f_1, \dots, f_n]$ for some $f_i \in k(Y)$. Take $0 \neq f \in k[Y]$ so that $ff_i \in k[Y]$. Then $k[X]_f = k[Y]_f$ and $D_X(f) \cong D_Y(f)$ via ϕ as varieties. So we just take $U = D_X(f)$. The other direction is clear since X, Y are irreducible and hence $U, \phi U$ are dense in X and Y respectively. $\square$

Theorem 4.1. *Let X, Y irreducible varieties and $\phi : X \rightarrow Y$ a dominant morphism. Set $r = \dim X - \dim Y$. Then there is a nonempty open subset U of X such that $\phi|_U$ is an open morphism, and if Y' is an irreducible closed subvariety of Y and X' is an irreducible component of $\phi^{-1}Y'$ intersecting U then $r = \dim X' - \dim Y'$. In particular, for $y \in Y$, the irreducible component of $\phi^{-1}(y)$ intersecting U has dimension r .*

Moreover, if $k(X)$ is algebraic over $k(Y)$ then for $x \in X$ the number of points in the fibre $\phi^{-1}(\phi(x))$ equals the separable degree $[k(X) : k(Y)]_s$

Proof. This result is built on three lemmas in [T.A10, section 5.1]. $\square$

For X, Y affine varieties, recall that $\phi : X \rightarrow Y$ is a **finite morphism** if $\phi^* : k[Y] \rightarrow k[X]$ makes $k[X]$ a f.g. module over $k[Y]$, or finite $k[Y]$ -algebra. We say ϕ is **locally finite** at $x \in X$ if there is a finite morphism $\mu : Y' \rightarrow Y$ and an isomorphism v of an open neighborhood U of x onto some open subset of Y' such that $\phi|_U = \mu \circ v$.

Note that finite morphism has finite fibres, i.e. is quasi-finite. To see this, it is known that finite morphisms are stable over base change, and the fibre over $y \in Y$ is essentially $X \times_Y \kappa(y)$. Hence it suffices to show that the spectrum of a finite k -algebra is a finite set, which is clear since a finite k -algebra is artinian.

In the exercises of [Har77], we showed that finite morphisms are closed, and that quasi-finiteness+properness $\Leftrightarrow$ finiteness.

Still let X, Y irreducible varieties and $\phi : X \rightarrow Y$ a dominant morphism. Write $A = k[Y]$, $B = k[X]$

Lemma 4.2. *Assume there is $b \in B$ such that $B = A[b]$. Let $x \in X$. Then either $\phi^{-1}(\phi x) \cong \mathbb{A}^1$ or $\phi^{-1}(\phi x)$ is finite and ϕ is locally finite at x .*

Proof. We have $B = A[T]/I$ where I is the ideal of functions vanishing at b . Let $\epsilon : A \rightarrow k$ be the homomorphism defining ϕx . Then $k[\phi^{-1}(\phi x)] = k[T]/\epsilon I \cdot k[T] = k[T]$ and hence $\phi^{-1}(\phi x) \cong \mathbb{A}^1$. Otherwise the polynomials in ϵI vanishes at $b(x)$, and so it contains nonconstant polynomials and no nonzero constant. It follows that $\phi^{-1}(\phi x)$ is finite, and there is $f \in I$ of the form $f_n T^n + \dots + f_0$ and $0 < m \leq n$ such that $\epsilon(f_i) = 0$ for $i > m$ and $\epsilon(f_m) \neq 0$. Set $s = f_n b^{n-m} + \dots + f_m$. Then $sb^m + f_{m-1}b^{m-1} + \dots + f_0 = 0$ and so sb is integral over $A[s]$ and b is integral over $A[s^{-1}]$. Since $s \in A[b]$, s is also integral over $A[s^{-1}]$ and hence over A . Now just notice that $B_s = A[sb, s]$ $\square$

Corollary 4.1. *Let $x \in X$. If $\phi^{-1}(\phi x)$ is finite then ϕ is locally finite at x . In particular, $\dim X = \dim Y$*

Proof. Write $B = A[b_1, \dots, b_n]$ and apply induction on n and lemma 4.2. See [T.A10, proposition 5.2.6] $\square$

Recall that a point x of an irreducible variety X is **normal** if the local ring $\mathcal{O}_{X,x}$ is integrally closed, or equivalently, there is an affine open neighborhood U of x such that $k[U]$ is integrally closed. Since regular local rings are UFD and hence integrally closed, smoothness implies normality.

Theorem 4.2. *(Zariski's main theorem) Let $\phi : X \rightarrow Y$ be a morphism of irreducible varieties that is bijective and birational and assume that Y is normal. Then ϕ is an isomorphism.*

Proof. Take $x \in X$. Replace X, Y with affine open neighborhoods U, V of $x, \phi(x)$ respectively. By lemma 4.2, ϕ is locally finite at x and so we may assume that U is isomorphic to an affine open subset of some affine variety V' that is finite over V . But birationality implies that $k(V') \cong k(V)$. Then we deduce from the normality of V that the finite morphism $V' \rightarrow V$ is in fact an isomorphism. This shows that ϕ is an isomorphism of ringed spaces, and so an isomorphism of varieties. $\square$

Proposition 4.1. *Let A be a normal integral domain with fractional field F . Let B be an integral domain that is a finite A -algebra. Assume that the fractional field E of B is separable over F . Then there is a nonzero element $a \in A$ such that the localization Ba is normal.*

Proof. See [T.A10, lemma 5.2.10] $\square$

Proposition 4.2. *Let X be an irreducible variety. The set of its normal points is nonempty and open.*

Proof. By definition clearly it is open. It is nonempty since we know there is a smooth point on the variety. $\square$

Now we give some applications involving algebraic groups with previous results. Let G be an algebraic group and X a homogeneous space for G . Let G^0 be the identity component of G . Let X' be an orbit of G^0 in X . Since G acts transitively on X , X is the disjoint union of the translates $g \cdot X'$ for $g \in G$. Each is a G^0 -orbit and is irreducible. Lemma 1.1 tells that these G^0 -orbits are closed. It follows that

Lemma 4.3. 1. *Each irreducible component of X is a homogeneous space for G^0*

2. *The components of X are open and closed and X is their disjoint union.*

Theorem 4.3. *Let $\phi : X \rightarrow Y$ be a G -equivariant homomorphism of homogeneous spaces for G . Put $r = \dim X - \dim Y$*

1. *For any variety Z , the morphism $(\phi, id) : X \times Z \rightarrow Y \times Z$ is open*

2. *If Y' is an irreducible closed subvariety of Y and X' an irreducible component of $\phi^{-1}Y'$ then $r = \dim X' - \dim Y'$. In particular, for $y \in Y$ the irreducible components of the fibre $\phi^{-1}y$ have dimension r .*

3. *ϕ is an isomorphism iff it is bijective and for some $x \in X$ the tangent map $d\phi_x : T_x X \rightarrow T_{\phi(x)} Y$ is bijective.*

Proof. Using lemma 4.3 it is reduced to the case that G is connected and X, Y are irreducible. Note that ϕ is surjective since G acts on X, Y transitively and ϕ is G -equivariant, and in particular, ϕ is dominant. Let U be the open subset of X with properties in theorem 4.1. Then all translates $g \cdot U$ have the same property. Since they cover X , the first assertions follow from theorem 4.1 immediately.

If ϕ is injective, the corollary of theorem 4.1 shows that $k(X)$ is purely inseparable over $k(Y)$. If $d\phi_x$ is surjective for some x , theorem 3.4 (2) also shows that ϕ is separable. We deduce that $k(X) = k(Y)$ and so ϕ is birational. Using theorem 4.2 and a covering argument, we conclude that ϕ is an isomorphism $\square$

The simplest example of bijective homomorphism of algebraic groups which fails to be an isomorphism is the Frobenius $x \mapsto x^p$ for $G = \mathbb{G}_a$ or $\mathbb{G}_m$ and $\text{char}(k) = p$. Here the differential $d\phi$ is the zero map at each point.

Corollary 4.2. *If $\phi : G \rightarrow G'$ is a surjective homomorphism of algebraic groups, then $\dim G = \dim G' + \dim \ker(\phi)$, and ϕ is an isomorphism iff ϕ is bijective and $d\phi_e$ is bijective*

Proof. This is immediate by viewing G, G' as homogeneous spaces for G . $\square$

Now let G be a connected linear algebraic group and σ an automorphism of G . Let G_σ be the set of fixed points of σ , which is a closed subgroup of G . Write χ for the morphism of G sending $x \mapsto \sigma(x)x^{-1}$. χ is then the composition

$$G \xrightarrow{(\sigma, id)} G \times G \xrightarrow{m \circ (id, i)} G$$

where m is the multiplication and i is the inverse. We deduce from the chain rule that $d\chi = d\sigma - 1$. Set $\mathfrak{g}_\sigma = \ker(d\chi)$. Since σ acts identically on G_σ , $L(G_\sigma)$ is fixed by $d\sigma$ and so $L(G_\sigma) \subset \ker(d\chi) = \mathfrak{g}_\sigma$. Note that χ defines a dominant G -equivariant morphism $\psi : G \rightarrow \chi G$. Also χG is an orbit in $\chi \overline{G}$ and so is dense open by lemma 1.1. Hence it contains a simple point, and so all points (in particular the identity e) are simple in χG . Also by theorem 4.3 (2) $\dim G_\sigma = \dim G - \dim \chi \overline{G}$ since G_σ is the kernel of ψ

Lemma 4.4. *ψ is separable iff $L(G_\sigma) = \mathfrak{g}_\sigma$*

Proof. Note that $\dim \text{Im}(d\psi_e) = \dim L(G) - \dim \mathfrak{g}_\sigma \leq \dim L(G) - \dim L(G_\sigma) = \dim G - \dim G_\sigma = \dim \chi \overline{G}$. Then it follows from theorem 3.4. $\square$

We say σ is semisimple if the induced automorphism σ^* of $k[G]$ is semisimple. Such automorphism has the following characterization

Theorem 4.4. *Assume σ is semisimple. Then χG is closed and $\psi : G \rightarrow \chi G$ is separable, and so $L(G_\sigma) = \mathfrak{g}_\sigma$*

Proof. Assume G is a closed subgroup of some GL_n and that $\sigma x = sxs^{-1}$ for some semisimple $s \in GL_n$, which we may take to be diagonal. We first show that ψ is separable. If $G = GL_n$, it can be checked easily. In general, we can extend σ to an automorphism of GL_n , which is conjugation by s , and similarly for χ . Take any $X \in T_e \chi \overline{G} \subset T_e \chi(GL_n)$. Since we know the result for GL_n , $d\chi$ is surjective and there is $Y \in \mathfrak{g}$ such that $X = d\chi(Y) = d\sigma(Y) - Y$. Semisimplicity of σ implies that $d\sigma$ is a semisimple automorphism of $\mathfrak{gl}_n$ stabilizing $\mathfrak{g}$. Then $\mathfrak{g}$ has a $d\sigma$ -stable complement. Hence we may take $Y \in \mathfrak{g}$, and $d\psi_e$ is surjective. Applying theorem 3.4 again we see that ψ is separable.

It remains to show that χG is closed. Put $f(T) = \prod (T - a_i)$ where a_i are distinct eigenvalues of s . Let S be the subset of GL_n consisting of matrices x such that x normalizes G , $f(x) = 0$ and the characteristic polynomial of the restriction of Ad_x to $\mathfrak{g}$ equals to that of $d\sigma$ (recall that Ad_x is

induced by the inner automorphism φ_x). Then S is closed and contains s . Since f has distinct roots, all elements of S are semi-simple.

Now set $G_x = \{g \in G | xgx^{-1} = g\}$ and $\mathfrak{g}_x = \{X \in \mathfrak{g} | Ad_x X = X\}$. By the first part and letting σ be the inner automorphism φ_x , it follows that $\dim G_x = \dim \mathfrak{g}_x$. The latter dimension equals the multiplicity of the eigenvalue 1 of the restriction of Ad_x to $\mathfrak{g}$, which equals $\dim \mathfrak{g}_\sigma$. Hence $\dim G_x = \dim G_\sigma$ for each $x \in G$. Now let G act on S by inner automorphisms. By theorem 4.3 (2), all orbits have dimension $\dim G - \dim G_\sigma$ (applied to the morphism $g \mapsto gxg^{-1}$ from G to the orbits). Also lemma 1.1 shows that all orbits are closed. Since χG is an orbit, we are done. $\square$

It follows that the conjugacy class $C = \{xsx^{-1} | x \in G\}$ in G for semisimple s is closed by letting σ be the inner automorphism. Let $Z = \{x \in G | xsx^{-1} = s\}$ be the centralizer of s . Then $C = s \cdot Im \chi$, $Z = G_\sigma$, and $\mathfrak{g} = (Ad_s - 1)\mathfrak{g} \oplus L(Z)$. In general, conjugacy classes in G are not closed. For example, the unique nontrivial conjugacy class of a unipotent element in SL_2 has the identity element in its closure. See [T.A10, exercise 5.4.9 (6)]

Finally we are ready to tackle the quotient group construction. Let H be a closed subgroup of the linear algebraic group G .

Lemma 4.5. *There is a f.d. subspace V of $k[G]$ and subspace W of V such that V is stable under right translations by elements of G , and $H = \{x \in G | \rho(x)W = W\}$ and $L(H) = \mathfrak{h} = \{X \in \mathfrak{g} | X.W \subset W\}$*

Proof. Let $I \subset k[G]$ be the defining ideal of H and V be a f.d. $\rho(G)$ -stable subspace of $k[G]$ containing a set of generators $f_1, \dots, f_r$ of I . Set $W = I \cap V$. We claim that W has the required property. Indeed, $H = \{x \in G | \rho(x)I = I\}$, hence $x \in H \implies \rho(x)W = W$. Conversely, if this is so, then $\rho(x)f_i \in I$ and so $\rho(x)I \subset I$ and $x \in H$. The proof for Lie algebra is similar. Note that if ϕ is the rational representation of G in V defined by ρ , then we have $d\phi(X).f = X.f$ for $f \in V, X \in \mathfrak{g}$ $\square$

Let V be an arbitrary f.d. vector space and W a subspace of dimension d . $\wedge^d V$ contains the $\dim-1$ subspace $L = \wedge^d W$. Let ϕ be the canonical representation of $GL(V)$ over $\wedge^d V$

Lemma 4.6. *For $x \in GL(V)$, we have $x \cdot W \subset W$ iff $\phi x(L) \subset L$, and for $X \in \mathfrak{gl}(V)$ $X \cdot W \subset W$ iff $d\phi X(L) \subset L$*

Proof. See [T.A10, lemma 5.5.2] $\square$

The lemmas imply the following result by taking V to be $\wedge^d V$ for V from lemma 4.5

Theorem 4.5. *There is a rational representation $\phi : G \rightarrow GL(V)$ and a nonzero $v \in V$ such that $H = \{x \in G | \phi x(v) \in kv\}$ and $\mathfrak{h} = \{X \in \mathfrak{g} | d\phi X(v) \in kv\}$*

Corollary 4.3. *There is a quasi-projective homogeneous space X for G , together with a point $x \in X$ such that the isotropy group of x in G is H , the morphism $\phi : g \mapsto gx$ from G to X defines a separable morphism $G^0 \rightarrow \phi G^0$, and the fibres of ϕ are cosets gH of H .*

Proof. Let V, v be in the theorem and x be the point in $\mathbb{P}(V)$ defined by the line kv . Denote by $\pi : V \setminus \{0\} \rightarrow \mathbb{P}(V)$ sending a vector to the line passing through it. G acts in an obvious way on $\mathbb{P}(V)$. Let X be the G -orbit of x . Lemma 1.1 shows that it is quasi-projective (open subvariety of some projective variety). The rest follows from the theorem and theorem 3.4. $\square$

Now we give the construction of the quotient space G/H once again. Let $\pi : G \rightarrow G/H$ be the canonical map. Define $U \subset G/H$ open if $\pi^{-1}U$ is open. This defines a topology on G/H that makes π open. Define a sheaf $\mathcal{O}$ of k -valued on G/H by declaring that for $U \subset G/H$ open $\mathcal{O}(U)$ is the ring of functions f on U such that $f \circ \pi$ is regular on $\pi^{-1}U$. Take X as in the previous corollary. One can

show that the map $\phi : G/H \rightarrow X$ sending $gH \mapsto gx$ is an isomorphism. See [T.A10, theorem 5.5.5]. By the corollary we get that G/H is a quasi-projective variety of dimension $\dim G - \dim H$, and if G is connected the canonical map π is separable.

In general, G/H is not affine. But this is so if H is normal in G and then G/H is a linear algebraic group as expected. See [T.A10, proposition 5.5.10]

More generally, for H a closed subgroup of a linear algebraic group G , let X be an irreducible H -variety on which H acts on right. We say the canonical map $\pi : G \rightarrow G/H$ has local sections if G/H can be covered by open U such that there are morphisms $\sigma : U \rightarrow G$ satisfying $\pi \circ \sigma = id_U$. Then the quotient $(G \times X)/H$ exist by gluing $U \times X$. It is denoted by $G \times^H X$. One has the morphism $G \times^H X \rightarrow G/H$ with local sections, whose fibers are isomorphic to X . So the quotient is the fibre bundle of G/H associated to X .

5 Parabolic and Borel subgroups

Recall that a variety X is **complete** if for any variety Y the projection $X \times Y \rightarrow Y$ is closed. This essentially says that the structure morphism $X \rightarrow \text{Spec } k$ is universally closed. By the notion of varieties, it is also separated and of finite type, and hence proper. Therefore a complete variety is the same as a proper k -variety

Example 6. The affine line $\mathbb{A}_k^1$ is not complete, since the projection map $\mathbb{A}^1 \times \mathbb{A}^1 \rightarrow \mathbb{A}^1$ mapping $V(xy - 1)$ onto $D(x)$

Proposition 5.1. *If X is complete, then*

1. *A closed subvariety of X is complete.*
2. *If Y is complete, so is $X \times Y$*
3. *If $\phi : X \rightarrow Y$ is a morphism of varieties, then ϕX is closed and complete*
4. *If X is a subvariety of some Y , then X is closed*
5. *If X is irreducible then any regular function on X is constant*
6. *If X is affine then X is finite.*

Proof. (1) and (2) follows from properties of properness. For (3), consider the graph $\Gamma = \{(x, \phi(x)) | x \in X\}$ of ϕ . By the separatedness of Y , this is a closed subvariety of $X \times Y$ isomorphic to X , and so is complete. Hence ϕX is closed since it is the image of Γ under the second projection. It is complete from the completeness of Γ and properties of properness. (4) follows from (3).

For (5), a regular function on X is a morphism $X \rightarrow \mathbb{A}^1$ that can be extended to a morphism $\phi : X \rightarrow \mathbb{P}^1$. If X is irreducible and ϕ is nonconstant, the image is a closed dense subset of $\mathbb{P}^1$ by (3) and so all of $\mathbb{P}^1$ (since the irreducible closed subset of $\mathbb{P}^1$ is either a singleton or all of $\mathbb{P}^1$), which is not possible. (6) follows from (5), since an affine variety of positive dimension has non-constant global function. $\square$

Theorem 5.1. *Any projective variety is complete*

Proof. Since projective varieties are closed subvariety of some $\mathbb{P}^n$, by proposition 5.1 (1) it suffices to show that $\mathbb{P}^n$ is complete, which is already shown in the algebraic geometry notes. $\square$

Theorem 5.2. *Let G be a connected algebraic group that is complete as a variety. Then G is commutative.*

Proof. See [T.A10, theorem 6.1.6] □

The groups in the theorem are essentially **abelian varieties**.

For the rest of this section, G is a linear algebraic group

Lemma 5.1. *Let X, Y be homogeneous spaces for G and $\phi : X \rightarrow Y$ a bijective G -morphism. Then X is complete iff Y is.*

Proof. By theorem 4.3 (1) for any variety Z the morphism $(\phi, id) : X \times Z \rightarrow Y \times Z$ is a homeomorphism. Hence ϕ is universally closed and so proper. The lemma follows immediately. □

We say a closed subgroup P of G is **parabolic** if the quotient variety G/P is complete. *Although it has been almost sixty years since Borel and Tits introduced this terminology, no one seems to know why they chose this word.*

Lemma 5.2. *If P is a parabolic subgroup then G/P is a projective variety.*

Proof. In the construction of quotient variety we have seen that G/P is quasi-projective. Now proposition 5.1 (4) shows that it is projective. □

Lemma 5.3. *If P is parabolic in G and Q is parabolic in P then Q is parabolic in G .*

Proof. See [T.A10, lemma 6.2.3] □

Lemma 5.4. *Let P be parabolic in G . If Q is a closed subgroup containing P , then Q is parabolic. Also P is parabolic in G iff P^0 is parabolic in G^0 .*

Proof. Since G/Q is the image of G/P under the canonical morphism, by proposition 5.1 (3) Q is parabolic. Note that G^0 is parabolic in G since G/G^0 is finite and discrete and hence complete. Hence if P is parabolic in G then P^0 is also parabolic in G by lemma 5.3. Since G^0/P^0 is closed in G/P^0 , it follows that P^0 is parabolic in G^0 . Conversely, if P^0 is parabolic in G^0 then it is also parabolic in G . Then P is parabolic in G by the first assertion □

Proposition 5.2. *A connected group G has proper parabolic subgroups if and only if it is not solvable (as an abstract group).*

Proof. Assume that G is a closed subgroup of some $GL(V)$. We have seen that G acts on the projective variety $\mathbb{P}(V)$. By lemma 1.1 we pick a closed orbit X for this action. Then X is complete, and we choose a point $x \in X$. Let P be its isotropic group. Then $gP \mapsto g \cdot x$ defines a bijective morphism of homogeneous spaces as in the discussion after corollary 4.3, and lemma 5.1 shows that G/P is complete and so P is parabolic in G . If $P = G$ put $V_1 = V/x$. Then G acts on $\mathbb{P}(V_1)$, and the same argument produces a parabolic subgroup P_1 . Continuing this way we either get a proper parabolic subgroup or realize G as a subgroup of some T_n , the upper triangular matrices. It is easy to see that any such subgroup is solvable.

It remains to show that if G is connected and solvable then it has no proper parabolic subgroup. Otherwise, assume P is one with maximal dimension. By lemma 5.4 we may assume P is connected. The commutator subgroup G_0 is closed and connected and $Q = P \cdot G_0$ is a connected parabolic subgroup

containing P . Then maximality forces $Q = P$ or $Q = G$. For the first case we have a bijection of homogeneous spaces for G_0

$$G_0/(G_0 \cap P) \rightarrow G/P$$

by lemma 5.1 it follows that $G_0 \cap P$ is parabolic in G_0 . Hence by applying induction on $\dim G$, we may assume that $G_0 = G_0 \cap P$, i.e. $G_0 \subset P$, which contradicts to the assumption that $Q = G$. For the case $Q = P$ we still have $G_0 \subset P$. But then P is normal in G and G/P is affine and so finite, contradiction. $\square$

Theorem 5.3. (*Borel's fixed point theorem*) *Let G be a connected solvable linear algebraic group and X a complete G -variety. There exists a point in X that is fixed by all elements of G .*

Proof. By lemma 1.1 G has a closed orbit in X , and we have seen in the above proof that the isotropic group of a point in that orbit is parabolic. By proposition 5.2 it has to be G , and hence the chosen point is as required $\square$

This motivates the following definition: a **Borel subgroup** of G is a closed connected solvable subgroup which is maximal for these properties. Such subgroups always exist and we take one with maximal dimension.

Theorem 5.4. 1. *A closed subgroup of G is parabolic if and only if it contains a Borel subgroup*
 2. *A Borel subgroup is parabolic;*
 3. *Two Borel subgroups of G are conjugate*

Proof. Again, by lemma 5.4 we may assume G is connected. Let B be a Borel subgroup and P any parabolic subgroup. Applying Borel's fixed point theorem to B and the complete variety G/P , we see that P contains a conjugate of B which is also a Borel subgroup. To prove the other direction of the first assertion by lemma 5.4 it suffices to prove (2). We may assume G is not solvable, and so it has a proper parabolic subgroup, which after conjugation we may assume contains B . Then B is Borel in P and by induction on $\dim G$ we have B parabolic in P , and so in G . For (3), say B, B' are two Borel subgroups. Then they are parabolic and each contains a conjugate of the other, whence they have the same dimension and both are conjugate. $\square$

Corollary 5.1. *Let $\phi : G \rightarrow G'$ be a surjective homomorphism of algebraic groups. Let P be a parabolic subgroup (resp. a Borel subgroup) of G . Then ϕP is a subgroup of G' of the same type.*

Proof. By theorem 5.4 (1) it suffices to deal with the case of Borel subgroups. Then ϕP is closed connected and solvable. Moreover, the induced morphism $G/P \rightarrow G'/\phi P$ is surjective. By proposition 5.1 ϕP is then parabolic and hence contains a Borel subgroup of G' . It follows that ϕP is Borel by maximality. $\square$

Recall we denote the center of G by $Z(G)$

Corollary 5.2. *Let B be a Borel subgroup of G . If G is connected then $Z(G)^0 \subset Z(B) \subset Z(G)$*

Proof. Note that $Z(G)^0$ is connected, closed and commutative (particularly solvable), and so lies in some Borel group. By the conjugacy of Borel groups, it in fact lies in all Borel groups, whence the first inclusion. For $g \in Z(B)$, $x \mapsto gxg^{-1}x^{-1}$ gives a morphism $G/B \rightarrow G$ and so must be constant and trivial by proposition 5.1 (5). Hence $g \in Z(G)$, which proves the second inclusion. $\square$

Corollary 5.3. *If B is nilpotent (as an abstract group) then $G^0 = B$.*

Proof. Note that a connected nilpotent group contains a non-trivial closed, connected group in its center (the subgroup generated by commutators of maximal length, see [T.A10, corollary 2.2.8 (1)]). Then by corollary 5.2 $Z(B)$ is nontrivial and central in G . Moduling $Z(B)$ and applying induction on $\dim G$ the result follows. $\square$

5.1 Connected Solvable Groups

In this section we give the structural theorem for connected solvable linear algebraic group. Across this section G is such a group. We will show that G always contains a torus and we will relate its structure to that of the torus.

Theorem 5.5. (*Lie-Kolchin Theorem*) Assume G is a closed subgroup of some GL_n . Then there is $x \in GL_n$ such that $xGx^{-1} \subset \mathbb{T}_n$, the group of upper triangle matrices.

In these notes we have used $\setminus$ to denote the upper triangle matrices following notations in the Lie algebra notes in the beginning part. Just a clarification.

Proof. Using induction on n it suffices to prove that the elements of G have a non-zero common eigenvector. This follows from Borel's fixed point theorem, applied to G acting on $\mathbb{P}(V) = \mathbb{P}^{n-1}$. Another elementary proof can be found in [T.A10, theorem 6.3.1] $\square$

As an interesting historical aside, Kolchin originally proved this result for differential Galois groups. These are groups attached to the solution spaces of linear homogeneous differential equations with function coefficients coming from a given field of functions, just as ordinary Galois groups are attached to the splitting fields of polynomials with coefficients from a given field of numbers. Differential Galois groups were historically the first groups to be called algebraic.

Corollary 5.4. Assume further that $G \subset GL_n$ is nilpotent. Then G_s and G_u are closed connected subgroups with G_s a central torus in G . Moreover, the product morphism $G_s \times G_u \rightarrow G$ is an isomorphism of algebraic groups.

Remark 5.1. We have seen the same result for commutative linear algebraic groups in theorem 2.1

Proof. First we show that G_s is the central torus of G . By nilpotency, $G_n = 0$ for some n where the G_n is the n -th group in lower power series. For s semisimple in G , let χ be the map defined by the inner automorphism by s and one has $\chi(x) = sxs^{-1}x^{-1} = [s, x]$ for $x \in G$. Whence $\chi^n G = e$. By [T.A10, lemma 4.4.13], the linear map $Ad_s - 1$ on $\mathfrak{g}$ is then both nilpotent and semisimple and so trivial. Hence G_s is closed under multiplication and is indeed a central torus. It follows that $V = k^n$ decomposes as a direct sum of one-dimensional subspaces, on each of which G_s acts by scalars. Lie-Kolchin then enables us to put the restriction of G to each subspace in triangular form. The result of the proof proceeds as in the commutative case $\square$

Corollary 5.5. Let G be connected solvable. Then the commutator subgroup $G_0 = [G, G]$ is closed, connected, unipotent, and normal. And G_u is a closed connected nilpotent, and normal subgroup of G ; the quotient group G/G_u is a torus

Proof. G_0 is connected and closed. By theorem 5.5 we may assume that G is a closed subgroups of some $\mathbb{T}_n$. Then it is clear that G_0 is unipotent. Since $G_u = G \cap U_n$, G_u is a closed normal subgroup, and nilpotent since U_n is. We have an injective map $G/G_u \hookrightarrow \mathbb{T}_n/U_n \cong D_n$. Thus G/G_u is diagonalizable, and is a torus since it is connected. It remains to show that G_u is connected. Since G_u^0 is normal in G , passing to G/G_u^0 it suffices to show that G_u is finite $\implies$ it is trivial. To see this, note that any

normal finite N of a connected linear algebraic group H is central, since for any $n \in N$, then morphism from G to N sending $g \mapsto gng^{-1}$ has connected and finite image which is then trivial. Hence $G_u = 1$ and $G = G_s$ is nilpotent. By corollary 5.4 we deduce that G_u is connected as required. $\square$

Lemma 5.5. *Assume that G is not a torus. There exists a closed normal subgroup N of G that is isomorphic to $\mathbb{G}_a$ and lies in the center of G_u*

Proof. We have seen in the proof of corollary 5.3 that we have a nontrivial closed connected subgroup H of G lying in the center of G_u . If $\text{char}(k) = p > 0$ we may assume that $H^p = e$ by replacing H with a suitable H^{p^m} . Then H is isomorphic to a vector group $\mathbb{G}_a^n$ by [T.A10, theorem 3.4.7]. If $n = 1$, we are done. Otherwise, let $k[H]$ be the space of additive functions on H . The torus $T = G/G_u$ acts on H by conjugation, and we obtain a representation of G/G_u over $k[H]$ with invariant subspace A . Further we can find $f \in A$ which is a simultaneous eigenvector for the T -action. Then $(\ker f)^0$ has the same property as H but with lower dimension. So we are done by induction. $\square$

5.2 Maximal Torus

A **maximal torus** of G is a subtorus of G that is not strictly contained in another subtorus. A **Cartan subgroup** of G is the identity component of the centralizer of a maximal torus.

The next result shows that a maximal torus of G in fact has the same dimension as $S = G/G_u$

Theorem 5.6. *Let G be connected solvable*

1. *Let $s \in G$ be semi-simple. Then s lies in a maximal torus. In particular, maximal tori exist*
2. *The centralizer $Z_G(s)$ of a semisimple element $s \in G$ is connected.*
3. *Any two maximal tori of G are conjugated.*
4. *If T is a maximal torus, then the product map $\pi : T \times G_u \rightarrow G$ is an isomorphism of varieties*

Proof. See [T.A10, theorem 6.3.5] $\square$

Corollary 5.6. *Let H be a subgroup with semisimple elements of connected solvable G . Then H is contained in a maximal torus of G ; in particular, any subtorus of H lies in a maximal torus. And $Z_G(H) = N_G(H)$ is connected*

Proof. Note that H is commutative since the restriction of the canonical map $G \rightarrow G/G_u$ to H is bijective. If H lies in the center of G then we are done. Otherwise, take a non-central element $s \in H$. By theorem 5.6 (2), $Z_G(s)$ is connected and contains H , and is a proper subgroup of G . Hence the first assertion and the connectedness of $Z_G(H)$ follows from induction on $\dim G$ to $Z_G(s)$.

If $x \in N_G(H)$, for $h \in H$ we have $xhx^{-1}h^{-1} \in H \cap G_0 \subset H \cap G_u = \{e\}$. Hence $x \in Z_G(H)$ as required $\square$

From now on G is an arbitrary connected linear algebraic group

Theorem 5.7. *Any two maximal tori of G are conjugated.*

Proof. Take any Borel subgroup B of G . A maximal torus is connected and solvable and so lies in a Borel group. By theorem 5.4 (3) T is conjugated to a subtorus of B , which must be a maximal torus for B . The theorem now follows from theorem 5.6 (3). $\square$

Proposition 5.3. *Let T be a maximal torus of G and $C = Z_G(T)^0$ the corresponding Cartan subgroup*

1. C is nilpotent and T is its unique maximal torus
2. There exists $t \in T$ lying in only finitely many conjugates of C .

Proof. Clearly C contains T as a central subgroup. A Borel subgroup B of C containing T also has T as a central subgroup and must be nilpotent since $B/T \cong B_u$ is nilpotent by theorem 5.6 (4). By corollary 5.3 we get $B = C$ and by corollary 5.4 $T = B_s$ is the unique maximal torus.

For (2), we need a claim

Claim: Let S be a subtorus of G . There exists $s \in S$ such that $Z_G(s) = Z_G(S)$

It suffices to prove the claim when $G = GL_n$. We may further assume that S consists of diagonal matrices. The diagonal entries define characters of S . Let $\chi_1, \dots, \chi_m$ be those. Then the elements $s \in S$ with $\chi_i(s) \neq \chi_j(s)$ for $i \neq j$ have the required property, and they form a dense open set in S

Now choose $t \in T$ with $Z_G(t) = Z_G(T)$. If t lies in a conjugate gCg^{-1} , then $g^{-1}tg \in T$ and $T \subset Z_G(g^{-1}tg) = g^{-1}Tg$. By maximality of T , we get $g \in N_G(T)$, but C is known to have finite index in this group, whence (2) follows $\square$

Lemma 5.6. *Let H be a closed subgroup of G and X the union of conjugates of H*

1. X contains a nonempty open subset in $\overline{X}$. If H is parabolic then X is closed
2. Assume that H has finite index in its normalizer N and that there exist elements of H lying in only finitely many conjugates of H . Then $\overline{X} = G$

Proof. Assume H to be connected. Consider $Y = \{(x, y) \in G \times G | x^{-1}yx \in H\}$. Then Y is a closed subgroup of $G \times G$ and isomorphic to $G \times H$ and thus irreducible. Note that $(x, y) \in Y \implies (xH, y) \subset Y$. It follows that $Y_1 = \{(xH, y) | x^{-1}yx \in H\}$ is an irreducible closed subgroup of $G/H \times G$. Since $X = \pi Y_1$, where π is the second projection, (1) follows from [T.A10, theorem 1.9.5] and proposition 5.1 (3).

Since the fibres of the projection $Y_1 \rightarrow G/H$ have dimension $\dim H$ it follows from theorem 4.3 (2) that $\dim G = \dim Y_1$. If $x \in H$ lies in only finitely many conjugates of H then $\pi^{-1}(x)$ is finite, since H has finite index in N . By corollary 4.1 we have $\dim \overline{X} = \dim Y_1 = \dim G$, as required. $\square$

Theorem 5.8. 1. *Every element in G lies in a Borel subgroup*

2. *Every semisimple element of G lies in a maximal torus.*

3. *The union of the Cartan subgroups of G contains a dense open subset.*

Proof. Let T be a maximal torus, $C = Z_G(T)^0$ the corresponding Cartan subgroup and B a Borel subgroup containing C which exists since C is connected and nilpotent by proposition 5.3. Also, since T is the unique torus of C , we have $N_G(T) = N_G(C)$. By corollary 2.2, C has finite index in its normalizer. By proposition 5.3 (2), we can apply lemma 5.6 (2) letting $H = C$, which proves (3). Next applying lemma 5.6 with $H = B$, we see that conjugates of B cover G and (1) follows. Finally, (2) follows from (1) and theorem 5.6 (1). $\square$

Corollary 5.7. *Let B be a Borel subgroup of G . Then the center $Z(B)$ of B coincides with $Z(G)$.*

Proof. For $x \in Z(G)$, x lies in a Borel group by theorem 5.8 (1), and hence in all of them by conjugacy of Borel groups. The other direction is proved in corollary 5.2. $\square$

Theorem 5.9. *Let S be a subtorus of G . Then the centralizer $Z_G(S)$ is connected. Moreover, if B is a Borel subgroup then $Z_G(S) \cap B$ is a Borel subgroup of $Z_G(S)$ and every Borel subgroup of $Z_G(S)$ arises this way.*

Proof. Set $Z = Z_G(S)$. Take any $g \in Z$ and let B be a Borel subgroup containing g . Put $X = \{xB \in G/B \mid x^{-1}gx \in B\}$. Then X is a closed variety of G/B and the fibre of g via the projection $Y_1 \rightarrow G$ as in the proof of lemma 5.6 with $H = B$. Then X is complete. Now S acts on X by left multiplication. By Borel's fixed point theorem, there is $xB \in X$ fixed by action of S , i.e. $x^{-1}Sx \subset B$. Hence there is a Borel subgroup containing both g and S . It follows from theorem 5.6 (2) and corollary 5.6 that $g \in Z^0$ and the first assertion follows.

For any Borel subgroup B . Then $Z \cap B$ is connected by corollary 5.6, and solvable. Now it suffices to show that $Z/Z \cap B$ is complete. Note that there is a bijective morphism from $Z/Z \cap B$ to the Z -orbit $Y = Z \cdot B$ in G/B . Since the map $G \rightarrow G/B$ is open by definition, it is enough to show that Y is closed.

As the image of $Z \times X$ under a morphism, Y is irreducible. Then $\bar{Y}$ is also irreducible, and hence connected. If $y \in Y$ then $y^{-1}Sy \subset B$. This also holds when $y \in \bar{Y}$. Consider the morphism $\bar{Y} \times S \rightarrow B/B_u$ sending $(y, s) \mapsto y^{-1}syB_u$. Applying proposition 2.2, we get that for $y \in \bar{Y}$, $y^{-1}sy \in sB_u$, so that $y^{-1}Sy$ is a maximal torus of sB_u . By the conjugacy of maximal tori of that group there is $z \in B_u$ s.t. $y^{-1}Sy = z^{-1}Sz$. Hence $y \in Z \cdot B = Y$ and Y is closed as desired. The other direction follows from conjugacy of Borel groups. $\square$

By contrast with solvable groups, centralizers of semisimple elements in general groups need not be connected.

Corollary 5.8. *Let T be a maximal torus of G . Then $C = Z_G(T)$ is a Cartan subgroup of G and any Borel subgroup of G containing T also contains C .*

Proof. By the previous theorem we know C is connected. Also recall that C is nilpotent. $\square$

Theorem 5.10. *For any Borel subgroup B of G , we have $B = N_G(B)$.*

Proof. We argue by induction on $\dim G$. The result is trivial if G is solvable. Otherwise, set $H = N_G(B)$ and let $x \in H$. Fix a maximal torus T contained in B . Then $x^{-1}Tx$ is another maximal torus contained in B . By conjugacy of maximal tori in B , we may assume that $x^{-1}Tx = T$. Consider the homomorphism $\varphi : t \mapsto txt^{-1}$ of T to itself. If the image is a proper subgroup of T , then $S = (\ker \varphi)^0$ is a non-trivial subtorus of T . Moreover, x lies $Z = Z_G(S)$ and normalizes the Borel subgroup $Z \cap B$ of Z . If $Z \neq G$ then by inductive hypothesis on Z we have $x \in B$; If $Z = G$ then S lies in the center of G . Passing to G/S , we may apply the inductive hypothesis again to get $x \in B$.

If the image of φ is all of T , choose ϕ, V and v as in theorem 4.5 for G/B , realising B as an isotropic group of the line kv inside some rational representation V of G . Then $\phi(B_u), \phi(T)$ fix v since B_u is unipotent and T lies in the commutator subgroup $[H, H]$. But this implies that ϕ induces a morphism from the complete variety G/B to the affine one V , which must be constant. Then G fixes v , and so $H = G$ by the settings in theorem 4.5 and B is normal in G . But then G/B is unipotent and hence G is solvable, forcing $G = H = B$. $\square$

Corollary 5.9. *Let T be a maximal torus of G and B a Borel subgroup containing T . The map $x \mapsto xBx^{-1}$ induces a bijection of $N_G(T)/Z_G(T)$ onto the set of Borel groups containing T .*

Proof. Surjectivity follows from conjugacy of maximal tori in B . Injectivity follows from theorem 5.10 and corollary 5.6, i.e. the normalizer and centralizer of a torus in a Borel subgroup coincides. $\square$

We provide more definitions to conclude this section. Write $\mathcal{B}$ the set of all Borel subgroups of an algebraic group. By theorem 5.10, the map $xB \mapsto xBx^{-1}$ defines a bijection G/B onto $\mathcal{B}$, and so we can define a structure of projective variety on $\mathcal{B}$. It is independent of the choice of B . We have thus defined $\mathcal{B}$ the **variety of Borel subgroups of G** .

More generally, fixing a parabolic subgroup P containing B , we can define a structure of projective variety $\mathcal{P}$ on the conjugates of P . Then we get a surjective morphism of homogeneous spaces $\mathcal{B} \rightarrow \mathcal{P}$ sending a Borel subgroup to the unique conjugate of P that contains it

If N, N' are normal subgroup of G then so is NN' . Hence there is a unique maximal closed connected normal solvable subgroup of G , called its **(solvable) radical** and denoted by $R(G)$. Similarly, there is a unique maximal closed connected normal unipotent subgroup of G , called its **unipotent radical** and denoted by $R_u(G)$. Note that $R(G)_u = R_u(G)$. We say G is **semisimple** if $R(G) = e$ and **reductive** if $R_u(G) = e$. It follows that $R(G)$ is the identity component of the intersection of all Borel subgroups.

The study of reductive groups will be the primary in the rest of the notes

6 Root system

In this chapter we introduce root systems which are combinatorial data associated to linear algebraic groups and play a crucial role in classification of some groups. We will be focusing on what is not covered in the Lie algebra notes and keep it brief in what we already know. Across this chapter G is connected linear algebraic group and T a maximal torus of G .

The **rank of G** is the dimension of T , which is independent of the choice of T since any two maximal tori are conjugated. The character group $X(T) = \text{Hom}_{\text{alg}}(T, k^*)$ is then isomorphic to $\mathbb{Z}^n$. The Lie algebra $\mathfrak{g}$ is a rational representation of T via the restriction of the adjoint representation. Then $\mathfrak{g}$ is a direct sum of $\dim - 1$ T -subrepresentation $\mathfrak{g}_\alpha$ for $\alpha \in X(T)$ such that $t \cdot v = \alpha(t)v$ for any $t \in T, v \in \mathfrak{g}_\alpha$. Such $\mathfrak{g}_\alpha$ are the **root spaces**, and the characters α such that $\mathfrak{g}_\alpha \neq 0$ are the **roots**. Denote the set of roots by R .

Lemma 6.1. *Let S be a subtorus of T . Then $Z_G(S) = Z_G(T)$ iff S is not contained in any of the subgroups $\ker \alpha$ for $\alpha \in R$*

Proof. See [T.A10, lemma 7.1.2] □

For $\alpha \in P$ denote by G_α the centralizer of the subtorus $\ker \alpha$ of T . This is a closed connected subgroup.

Lemma 6.2. *G_α generate G as α runs over P . If all G_α are solvable then so is G*

Proof. The group H generated by G_α is closed and connected. Its Lie algebra contains the Lie algebra of the centralizer of T and all the root space $\mathfrak{g}_\alpha$, and hence all of $\mathfrak{g}$. So $\mathfrak{h} = \mathfrak{g}$, and $H = G$ by connectedness

If G_α is solvable then it lies in some Borel subgroup and so every Borel subgroup by theorem 5.9. If this holds for each $\alpha \in P$ then G is clearly solvable. □

We denote P' the set of roots α such that G_α non-solvable and W the quotient $N_G(T)/Z_G(T)$, called the **Weyl group** of G . We have seen that W is finite. It acts faithfully as a group of automorphisms of $X(T)$ permuting P and P' . By corollary 5.9 there is a bijection between W and Borel subgroups containing T . If B is such a group, then there is also a bijection between W and the T -fixed points of in G/B .

Note that for $\alpha \in P'$, G_α contains $S = \ker \alpha$ in its center and the Weyl group of G_α relative to T (notice that $T \subset G_\alpha$) coincides with G_α/S relative to T/S which is isomorphic to $\mathbb{G}_m$

Proposition 6.1. *Assume G is non-solvable and $\dim T = 1$. Then W has order two. If B is a Borel subgroup then $\dim G/B = 1$*

Proof. First note that W has order at most two since it acts faithfully on $X = X(T) \cong \mathbb{Z}$ which only has two automorphisms. Now fix an isomorphism $\lambda : T \xrightarrow{\sim} \mathbb{G}_m$. Let B be a Borel subgroup containing T . Let $\phi : G \rightarrow GL(V)$, $v \in V$ be as in theorem 4.5 for $H = B$. We may assume that V is spanned by the images $\phi(x)v$ for $x \in G$. The ϕ defines an isomorphism from G/B onto a closed subvariety $Y = G \cdot kv$ of $\mathbb{P}(V)$ by definition, and we identify G/B with this subvariety. Choose a basis $\{e_i\}$ of V of weight vectors for the induced representation ρ of $T \cong \mathbb{G}_m$ on it so that we have $\rho(a)e_i = a^{m_i}e_i$ for $a \in k^*$.

Assume that $m_1 > \dots > m_n$. For $x \in V \setminus \{0\}$ $x^* = kx \in \mathbb{P}(V)$ in homogeneous coordinates $(x_1, \dots, x_n)$. If i, j are respectively the largest and smallest indices with $x_i, x_j \neq 0$, then an easy argument shows that $ke_i, ke_j \in Y$. The morphism $\mathbb{G}_m \rightarrow \mathbb{P}(V)$ arising from the action of T extends in two ways to a map from $\mathbb{A}^1$ to $\mathbb{P}(V)$, with the extra point in its image lying in Y in both cases. One should think of ke_i as $\lim_{a \rightarrow \infty} \rho(a)(kx)$ and ke_j as $\lim_{a \rightarrow 0} \rho(a)(kx)$. Clearly they are both fixed by the action of $T \cong k^*$, and they are distinct, otherwise all points of $\mathbb{P}(V)$ will be fixed by T . Then the above observation of Weyl group implies that W has order two.

Then ke_i, ke_j are the T -fixed points in G/B . We find that the points in Y with j -th coordinate 0 form a closed T -stable subset Σ which is nonempty. If any component of Σ has dimension ≥ 1 , then the above argument would show that that component would include at least two fixed points of T , together with the one ke_j that it already has, a contradiction. Hence Σ is finite. The j -th coordinate function f on a suitable open neighborhood of e_i takes the value 0 there and $f^{-1}(0)$ is finite. The second assertion now follows from corollary 4.1 since $\dim(G/B)$ cannot be 0 $\square$

Now return to the general case. Let $\alpha \in P'$. It follows immediately from proposition 6.1 that $W_\alpha = W(G_\alpha, T)$ has order two. Choose $n_\alpha \in N_{G_\alpha}(T) \setminus Z_{G_\alpha}(T)$ and let s_α be the image of n_α in W . Denote by $X^\vee$ the group $\text{Hom}(X = X(T), \mathbb{Z})$ of cocharacters. Then there is a non-degenerating pairing $\langle \cdot, \cdot \rangle$ between X and $X^\vee$. Identify $X, X^\vee$ with the subgroups of $V = \mathbb{R} \otimes X$ and $V^\vee = \mathbb{R} \otimes X^\vee$. The induced pairing on $V, V^\vee$ is also denoted by $\langle \cdot, \cdot \rangle$. The action of W on X naturally extends to a linear action on V .

Now we introduce a positive definite symmetric bilinear form $(\cdot, \cdot)$ on V that is invariant under W . For any symmetric positive definite bilinear form f on V , replace $f(x, y)$ by $(x, y) = \sum_{w \in W} f(wx, wy)$. Then s_α are reflections in the Euclidean space V . It is known that $s_\alpha(v) = v - \frac{2(v, \alpha)}{(\alpha, \alpha)}\alpha$. Identify $V^\vee$ with V via the form $(\cdot, \cdot)$. Then set $\alpha^\vee = 2\alpha/(\alpha, \alpha) \in V^\vee$, and $s_\alpha(v) = v - \langle v, \alpha^\vee \rangle \alpha$

Theorem 6.1. W is generated by s_α for $\alpha \in P'$

Proof. See [T.A10, theorem 7.1.9] $\square$

One main goal of this section is to attach to $(G, \mathfrak{g})$ a **root datum**. This is a quadruple $(X, R, X^\vee, R^\vee)$, where $X, X^\vee$ are free abelian groups of finite rank, in duality by a pairing $\langle \cdot, \cdot \rangle : X \times X^\vee \rightarrow \mathbb{Z}$. $R, R^\vee$ are finite subsets of $V, V^\vee$, with a bijection $\alpha \mapsto \alpha^\vee$ between them. Then we have $\langle \alpha, \alpha^\vee \rangle = 2$ and $s_\alpha R = R, s_{\alpha^\vee} R^\vee = R^\vee$. In our settings, X is the character group of a maximal torus of G , R the set of its roots, $X^\vee$ the cocharacter group, and the remaining notations are as defined above. The root datum is independent of the choice of maximal torus T since any two such are conjugate.

The Weyl group of a root datum is the group of automorphisms of X generated by s_α for $\alpha \in R$. Denote by Q the subgroup of X generated by R and put $V' = \mathbb{R} \otimes Q$. Then R is a **root system** in V' , i.e. R is finite and V' is $\mathbb{R}$ -spanned by R ; $s_\alpha(R) \subset R$ for all $\alpha \in R$; $\langle \alpha, \beta^\vee \rangle \in \mathbb{Z}$ for all $\alpha, \beta \in R$

Before getting into the classification of root systems and root data, we need more lemmas of non-solvable groups of rank 1. Fix a Borel subgroup B containing a maximal torus T , let $U = B_u$ be its unipotent radical and let $n \in N_G(T)$ represent the nontrivial element of the Weyl group W as in proposition 6.1. Then $ntn^{-1} = t^{-1}$ and $n^2 \in Z_G(T)$

Lemma 6.3. 1. G is the disjoint union of B and UnB

2. $R(G) = (U \cap nUn^{-1})^0$

3. $\dim U / (U \cap nUn^{-1}) = 1$

Proof. Let $x \in G/B$ be the coset B . By the proof of proposition 6.1, $x, n \cdot x$ are the two T -fixed points in G/B . If $n \in N_G(B) = B$, then $n \in N_B(T) = Z_B(T) \subset Z_G(T)$ by theorem 5.6. Hence $n^{-1}Bn \neq B$ and $Un \cdot x \neq \{n \cdot x\}$. We have seen that $\dim G/B = 1$. It follows that the complement of $Un \cdot x$ is a finite set S . Since the torus T normalizes U , it must permute this set S , whence it fixes all of its points. It follows that $S \subset \{x, n \cdot x\}$. Since $x \notin Un \cdot x, n \cdot x \in Un \cdot x$ we conclude that $Un \cdot x = G/B - \{x\}$ and the first assertion follows.

Note that $U \cap nUn^{-1}$ is essentially the isotropic group of $n \cdot x$ in U . Then (3) follows from (1) and theorem 4.3 (2). Since the normalizer of a proper closed connected subgroup of a unipotent group always has larger dimension than the subgroup ([T.A10, exercise 6.3.7 (4)]), $(U \cap nUn^{-1})^0$ is normal in U . Since this group is also normalized by T and n , it is normal in G . Now (2) follows because $R(G)$ cannot contain a torus. □

Lemma 6.4. Assume that G is connected semisimple of rank one

1. $\dim U = 1$, $Z_G(T) = T$ and $U \cap nUn^{-1} = e$

2. There is a unique weight α of T in $\mathfrak{g}$ such that $\mathfrak{g}$ is the direct sum of $\mathfrak{t} = L(T)$ and two one dimensional weight spaces $\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}$ with $L(U) = \mathfrak{g}_\alpha, L(nUn^{-1}) = \mathfrak{g}_{-\alpha}$

3. The product map $(u, b) \mapsto unb$ is an isomorphism of varieties $U \times B \rightarrow UnB = G - B$

Proof. From lemma 5.3 we see that $\dim U = 1$ and $U \cap nUn^{-1}$ is finite and unipotent. Since this group is normalized by T , it lies in the centralizer of T , which is connected and contained in B . Also theorem 5.6 (4) shows that $\dim B = 2$. Hence $Z_G(T) = T$ or $Z_G(T) = B$. The second would force B to be nilpotent and G solvable, a contradiction. Hence $Z_G(T) = T$ and $U \cap nUn^{-1} = e$

By theorem 2.3 we see that $U \cong \mathbb{G}_a$. Let $u : \mathbb{G}_a \rightarrow U$ be an isomorphism. There is a character α of T such that $tu(a)t^{-1} = u(\alpha(t)a)$ for $t \in T, a \in k$. And α is nontrivial since $Z_G(T) = T$. If $X \in L(U)$ is a nonzero element in the image of the differential du , then for $t \in T$, $Ad_t X = \alpha(t)X$, $Ad_n X \in L(nUn^{-1})$, and $Ad_t Ad_n X = \alpha(t)^{-1} Ad_n X$

From lemma 6.3 (1) we see that $\dim G \leq 3$. But it follows from above that $\mathfrak{t} \oplus kX \oplus kAd_n X$ is a subspace of $\mathfrak{g}$. Hence $\dim G = 3$ and (2) follows.

For (3), it suffices to show that $(v, b) \mapsto vb$ is an isomorphism of the group $nUn^{-1} \times B$ onto $G - nB$. Using (2), we see that the tangent map at (e, e) of this map is bijective. Since we can view this as an equivariant map of homogeneous spaces, now it follows from theorem 4.3 (3) □

Theorem 6.2. The only connected semisimple groups of rank one up to isomorphism are $SL_2(k)$ and $PSL_2(k)$

Proof. We have done most of the pre-work, but some case-by-case discussion is still needed. See [T.A10, theorem 7.2.4] □

We now study reduced (crystallographic) root systems in Euclidean space. We say a root system (V, R) is **reduced** if for $\alpha \in R, c \in \mathbb{Q}, c\alpha \in R \iff c = \pm 1$. The root systems arising from reductive groups are reduced, since we notice that $G_\alpha = G_{c\alpha}$ and the pair of roots $\{\pm\alpha\}$ is uniquely determined by G_α .

For $\alpha, \beta \in R$, note that $\frac{4(\alpha, \beta)^2}{(\alpha, \alpha)(\beta, \beta)} \in \mathbb{N}$ and by Cauchy-Schwartz it can only take values 0, 1, 2, 3. Hence the angle between any two non-proportional roots is necessarily a multiple of $\pi/4$ or $\pi/6$. This is what makes R crystallographic. From this we can easily deduce that the root systems of rank 2 are of type $A_1 \times A_1, A_2, B_2$ (or C_2) or G_2 up to isomorphism.

Recall that a Weyl chamber is a connected component of the complement in V of the union of hyperplanes H_α orthogonal to α . Given x lying in a Weyl chamber, we say a root α is **positive** (relative to x) if $(x, \alpha) > 0$. The set R^+ of positive roots is called a **positive subsystem**. We say a positive root is **simple** if it cannot be written as the sum of two positive roots.

In the Lie algebra notes we have seen that the structure of the positive system is independent of the choice of the Weyl chamber. Also every root is an integer combination of simple roots, with all coefficients nonnegative or all coefficients nonpositive.

We conclude by relating positive subsystems to Borel subgroups of linear algebraic groups

Lemma 6.5. *Let G be a connected semisimple group of rank 1, B a Borel subgroup, T a maximal torus of B , and α the unique root of T in B . Let χ be a character of T , regarded as a character of B via the isomorphism $B/B_u \cong T$. Assume that $f \in k[G]$ is a non-constant regular function such that for $g \in G, b \in B$ we have $f(gb) = \chi(b)f(g)$. Then $\langle \chi, \alpha^\vee \rangle > 0$*

Proof. By theorem 6.2 G is either SL_2 or PSL_2 . Since $k[PSL_2] \subset k[SL_2]$, we may assume that $G = SL_2$. Then we can take T to be the diagonal matrices and B the upper triangular matrices in G . Then $\alpha = e_1 - e_2$ and one checks that $\alpha(\text{diag}(x, x^{-1})) = x^2$ and $\alpha^\vee(x) = \text{diag}(x, x^{-1})$. Set $\langle \chi, \alpha^\vee \rangle = a$.

By the proof of [T.A10, theorem 7.2.4], we have for $z \neq 0$ $f\left(\begin{pmatrix} 1 & 0 \\ z & 1 \end{pmatrix}\right) = z^a g(z^{-1})$ for some polynomial g . Since f is regular at $z = 0$, we must have $a \geq 0$. If $a = 0$ then g must be constant and so is f . Hence $a > 0$ as required. $\square$

Now let G be an arbitrary connected algebraic group with Borel subgroup B and maximal torus $T \subset B$. Let α be a root of T in G . Then $G_\alpha \cap B$ is a Borel subgroup in G_α , whence $B' = (G_\alpha \cap B)/(R_u(G_\alpha) \cap B)$ is a Borel subgroup of the reductive group $G' = G_\alpha/R_u(G_\alpha)$, containing the image T' of T . Let $\pm\alpha'$ be the character of T' corresponding to $\pm\alpha$. It is easy to check that $L(B')$ is the direct sum of $L(T')$ and a one-dimensional weight space, whose weight is either α' or $-\alpha'$. It follows that B picks out one root from each pair of roots $\pm\alpha$. Let $R^+(B)$ be the set of roots obtained in this way as α runs through set $R(G, T)$ of roots of T in G .

Proposition 6.2. *With above notations, $R^+(B)$ is a positive subsystem.*

Proof. Choose a rational representation $\phi : G \rightarrow GL(A)$ with a nonzero vector $a \in A$ such that B is the stabilizer of the line ka as in theorem 4.5. Then there is a character χ of T , extended to B as above, with $(\phi b)(a) = \chi(b)a$. Let ℓ be a linear function on A and put $F(g) = \ell((\phi g)(a))$ for $g \in G$. Then $F \in k[G]$ and for $b \in B$ we have $F(gb) = \chi(b)F(g)$. Let α be a root and restrict F to G_α . Since the unipotent radical $R_u(G_\alpha)$ fixes a , the function F is the pullback of a function $F' \in k[G']$ so that F has the property of the hypothesis of lemma 6.5. Hence $\langle \chi, \alpha^\vee \rangle > 0$. It follows that $(R^+(B))^\vee$ is a positive subsystem of $R^\vee$. Hence $R^+(B)$ is a subsystem of positive roots of R . $\square$

In particular, if α, β are two roots of T in B such that $\alpha + \beta$ is a root, then it is a root of T in B . Also note that since any two positive subsystems are conjugate under the Weyl group W , any positive subsystem has the form $R^+(B)$ for a unique Borel subgroup B containing the fixed maximal torus T .

For a detailed classification of root systems and their Dynkin diagrams, see the Lie algebra notes.

7 Reductive Groups

We need a few more results on the unipotent radical

Theorem 7.1. *Let G be a connected group and T a maximal torus. Then $R_u(G)$ is the identity component of the intersection of the unipotent parts B_u of the Borel subgroup B containing T*

Proof. See [T.A10, theorem 7.6.3] □

Corollary 7.1. *Assume G to be reductive*

1. *Let $S \subset G$ be a subtorus. Then $Z_G(S)$ is connected and reductive*
2. *$Z_G(T) = T$, i.e. Cartan subgroups are maximal tori*
3. *The center $Z(G)$ of G lies in T*

Proof. (1) follows from theorem 7.1 and theorem 5.9. (2) follows from (1) by letting $S = T$ and that $Z_G(T)$ is nilpotent and hence its unipotent part is precisely the unipotent radical. For (3) just observe that $Z(G) \subset Z_G(T) = T$ □

From now on, G is a connected reductive linear algebraic group and T is a maximal torus of G . By corollary 7.1, the Weyl group of (G, T) is essentially $G = N_G(T)/T$

Proposition 7.1. *For $\alpha \in R$ there is an isomorphism u_α from $\mathbb{G}_a$ onto a unique closed subgroup U_α of G such that $tu_\alpha(x)t^{-1} = u_\alpha(\alpha(t)x)$ for $t \in T, x \in k$. Moreover, U_α and T generate G*

Proof. This is by [T.A10, lemma 7.3.3], and U_α is essentially the image of u_α defined there □

Proposition 7.2. *The center $Z(G)$ is the intersection of $\ker \alpha$ for roots α*

Proof. Then the statement is clear since $Z(G) \subset T$ following proposition 7.1 □

Consequently, $Z(G)$ is nontrivial iff the rank of the character group X of T is greater than the rank of R

For a root datum $(X, R, X^\vee, R^\vee)$, there is more than just the root system R . Let Q be the subgroup of X generated by R . This is the **root lattice**. Put $P = \{x \in V \mid \langle x, R^\vee \rangle \in \mathbb{Z}\}$. This is the **weight lattice**. Then $Q \subset P$. Assume G is semisimple. Then X, P, Q have the same rank. The finite quotient P/Q is the **fundamental group** of R .

In fact, there are only finitely many semisimple algebraic groups up to isomorphism with a fixed root system; more precisely, the isomorphism classes of such groups are in bijection to the lattices between Q and P . All have isomorphic Lie algebras.

Example 7. Recall that the **fundamental weights** $\{w_i\}$ are the dual basis of the simple roots w.r.t. the pairing, i.e. $\langle w_i, \alpha_j^\vee \rangle = \delta_{ij}$. Then P is essentially the $\mathbb{Z}$ -span of fundamental weights

In the Lie algebra notes we have computed that for A_{n-1} the fundamental weights are $w_k = e_1 + \dots + e_k - \frac{k}{n}(e_1 + \dots + e_n)$ for $k = 1, \dots, n-1$. Also note that in general, for a rank n root system $w_i = \sum_{j=1}^n (A^{-1})_{ij} \alpha_j$ where A is the Cartan matrix. Hence the index $[P : Q]$ is precisely the determinant of A . In particular, $|P(A_{n-1})/Q(A_{n-1})| = n$. Moreover, note that $nw_1 = (n-1)\alpha_1 + (n-2)\alpha_2 + \dots + \alpha_{n-1} \in Q(A_{n-1})$, and that $w_k - kw_1 \in Q(A_{n-1})$, i.e. $\overline{w_1}$ generates $P(A_{n-1})/Q(A_{n-1})$. Hence the fundamental group of A_{n-1} is cyclic of order n .

Similarly, we compute that the fundamental groups of B_n and C_n are cyclic of order 2; the fundamental group of D_n is cyclic of order 4 if n is odd but the direct product of two cyclic groups of order 2

if n is even. (To remember which is which, recall that the root systems A_3, D_3 are isomorphic, having the same Dynkin diagram.) In the exceptional cases, only types E_6 and E_7 have nontrivial fundamental groups. These have order 3 and 2 respectively.

We have seen that the tori are exactly the algebraic groups G such that the character group $X^{(G)}$ is free abelian of rank equal to the dimension of G and whose elements span the coordinate ring $k[G]$ over k . More precisely, there is an anti-equivalence of categories between tori and free abelian groups of finite rank. Using this fact and taking for granted that given any abstract root datum D there is a reductive group G with root datum D , unique up to isomorphism, we see that for a fixed root system R the inclusion $Q \subset P$ of lattices corresponds to a pair of tori T, T' such that there is a surjective map $T \rightarrow T'$ with finite kernel such that $X^*(P) = T, X^*(Q) = T'$

Passing to the algebraic groups G, G corresponding to the data $(P, R, P^\vee, R^\vee), (Q, R, Q^\vee, R^\vee)$ respectively, we deduce that there is a surjective homomorphism from G to G with finite central kernel. More generally, for any lattice L between Q and P , there is a surjective homomorphism from G to the algebraic group corresponding to $(L, R, L^\vee, R^\vee)$ with finite central kernel (whose order equals the index of L in P). The group corresponding to the largest choice P for X is said to be **simply connected**. The group corresponding to the smallest choice Q is said to be **adjoint, or of adjoint type**. It is the image $\rho(G)GL(\mathfrak{g})$ of any G with Lie algebra $\mathfrak{g}$ in the adjoint representation ρ of G on $\mathfrak{g}$. Any two such groups G are said to be **isogenous**. A surjective homomorphism from one of them to another with finite kernel is called an **isogeny**.

Example 8. For type A_{n-1} , the simply connected group is SL_n , and hence the other groups are the quotients of this group by a finite central subgroup of order dividing $|P(A_{n-1})/Q(A_{n-1})| = n$. The adjoint group is $PSL_n = SL_n/Z$ where Z is the center of SL_n generated by the scalar matrix $e^{2\pi i/n}I$.

For types B and D the adjoint group is SO_n . The simply connected group is denoted $Spin_n$ which is a subgroup of the Clifford algebra (see the Lie algebra notes) and a double cover of SO_n .

For type C_n the simply connected group is Sp_{2n} and the adjoint group is denoted PSp_{2n} .

Corollary 7.2. The roots of R are the non-zero weights of T in $\mathfrak{g}$. For each $\alpha \in R$ the weight space $\dim \mathfrak{g}_\alpha$ has dimension one.

Corollary 7.3. Let B be a Borel subgroup of G containing T and let $\alpha \in R$. Then

1. $\alpha \in R^+(B) \iff U_\alpha \subset B$ where U_α is given in proposition 7.1 $\iff \mathfrak{g}_\alpha \subset \mathfrak{b}$
2. $\dim B = r + \frac{1}{2}|R|$ where r is the rank of G , and $\dim G = r + |R|$

Proof. (1) follows from the definition of $R^+(B)$ in proposition 6.2 and that the $L(U_\alpha) = \mathfrak{g}_\alpha$. (2) follows from corollary 7.2 by calculating $\dim \mathfrak{g}$ and $\dim \mathfrak{b}$ $\square$

Proposition 7.3. 1. u_α in proposition 7.1 can be chosen such that $n_\alpha = u_\alpha(1)u_{-\alpha}(-1)u_\alpha(1)$ lies in $N_T(G)$ and its image in W is s_α . For $x \in k^*$ we have

$$u_\alpha(x)u_{-\alpha}(x^{-1})u_\alpha(x) = \alpha^\vee(x)n_\alpha$$

2. $n_\alpha^2 = \alpha^\vee(-1)$ and $n_{-\alpha} = n_\alpha^{-1}$
3. For $u \in U_\alpha \setminus \{1\}$ there is a unique $u' \in U_{-\alpha} \setminus \{1\}$ such that $uu'u \in N_G(T)$
4. If $\{u'_\alpha\}$ is a second family with the property above then there are $c_\alpha \in k^*$ s.t. $u'_\alpha(x) = u_\alpha(c_\alpha x), c_\alpha c_{-\alpha} = 1$ for $x \in k, \alpha \in R$

Proof. We have $U_\alpha \subset [G_\alpha, G_\alpha]$ which is a semisimple group of rank one, and hence isomorphic to SL_2 or PSL_2 by theorem 6.2. Therefore, it is reduced to the case $G = SL_2$ (this is an analogy to finding $(\mathfrak{sl}_2)_\alpha$ in Lie algebra). Then we may take $u_\alpha(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$, $u_{-\alpha} = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}$, $n_\alpha = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The statement can be checked in SL_2 easily. $\square$

We call the family $\{u_\alpha | \alpha \in R\}$ with the properties above **a realization of the root system** $R = R(G, T)$ in G . Note that the realization determines the coroots $\alpha^\vee$.

Finally we show that the variety structure of a Borel subgroup is as simple as one could hope for. Fix an ordering $(\alpha_1, \dots, \alpha_m)$ of the positive subsystem $R^+(B)$ corresponding to a Borel subgroup B and a realization $\{u_\alpha\}$ of $R = \pm R^+(B)$.

Proposition 7.4. *The morphism $\phi : \mathbb{G}_a^m \rightarrow B_u$ with $\phi(x_1, \dots, x_m) = u_{\alpha_1}(x_1) \dots u_{\alpha_m}(x_m)$ is an isomorphism of varieties. In particular, B_u is generated by the U_α with $\alpha \in R^+$*

The proposition follows from the next lemma:

Lemma 7.1. *Let H be a connected solvable algebraic group with maximal torus S . Assume that there is a set of isomorphisms v_i ($1 \leq i \leq n$) of $\mathbb{G}_a$ onto closed subgroups of H such that there exist nontrivial characters β_i of S , no two of which are linearly dependent with $sv_i(x)s^{-1} = v_i(\beta_i(s)x)$ for $s \in S, x \in k$, and the weight spaces $\mathfrak{h}_{\beta_i}$ are one-dimensional and span $L(H_u)$. Then the morphism $\psi : \mathbb{G}_a^n \rightarrow H_u$ with $\psi(x_1, \dots, x_n) = v_1(x_1) \dots v_n(x_n)$ is an isomorphism of varieties.*

Proof. We apply induction on n . If $n = 1$ then H_u is the image of v_1 by computing dimensions and the result is trivial. For $n > 1$, let N be a normal subgroup in the center of H_u isomorphic to $\mathbb{G}_a$ by lemma 5.5. Then $L(N)$ is a $\dim - 1$ S -subspace of $L(H_u)$, which must be one of the weight space $\mathfrak{h}_{\beta_i}$. Then the centralizer $Z_H(\ker \beta_j^0)$ is a group with the properties of H and has $\dim - 1$ unipotent radical (here uses the linear independence of β_j). By what we establish in $n = 1$ we see that N is the image of v_i .

For $i \neq j$ let $w_j : \mathbb{G}_a \rightarrow H/N$ be the homomorphism induced by v_j . We claim that w_j and H/N satisfy the assumption with the image of S in H/N the maximal torus. This is clear except for that w_j are isomorphisms. Note that the images of v_i and v_j have trivial intersection (otherwise, β_i would be a multiple of β_j), w_j is injective. By corollary 4.2, w_j is an isomorphism onto a closed subgroup of H/N and the claim follows. By inductive hypothesis on H/N , we see that ψ is bijective since N is central. Also it is clear that the tangent map $d\psi_{(0, \dots, 0)}$ is bijective. By theorem 4.1, ψ is birational. By Zarisk's main theorem, we see that ψ is an isomorphism $\square$

Naturally one would expect that the same thing holds after replacing H and $R^+(B)$ with G and R after extending the above ordering to R . However, this is not the case. Instead the image of morphism as defined in proposition 7.4 is a proper open subset of G . For now we introduce the structure constants that will play a crucial role in presenting a linear algebraic group with specified root datum as an abstract group.

Proposition 7.5. *Let $\alpha, \beta \in R$ with $\alpha \neq \beta$. There exist constants $c_{\alpha, \beta, i, j} \in k$ such that the commutator*

$$[u_\alpha(x), u_\beta(y)] = \prod_{i\alpha + j\beta \in R, i, j > 0} u_{i\alpha + j\beta}(c_{\alpha, \beta, i, j} x^i y^j)$$

for all $x, y \in k$, where the order of the factors on the right side is the one prescribed by the ordering of R . In particular, if there are no $i, j > 0$ such that $i\alpha + j\beta \in R$, then $u_\alpha(x)$ commutes with $u_\beta(y)$ for all x, y

Proof. It suffices to prove this for $\alpha, \beta \in R^+$. Then $U_\alpha, U_\beta \subset B_u$ and $[u_\alpha(x), u_\beta(y)] = \prod_{\gamma \in R^+} u_\gamma(P_\gamma(x, y))$ for some $P_\gamma \in k[U, T]$, where the order of factors in the product is the prescribed one. Conjugating by t by proposition 7.1 we get $P_\gamma(\alpha(t)x, \beta(t)y) = \gamma(t)P_\gamma(x, y)$. Using the linear independence of characters we deduce that $P_\gamma \neq 0 \iff \gamma = i\alpha + j\beta$ for some $i, j \geq 0$ and that P_γ is a multiple of $x^i y^j$. It remains to show that neither i nor j can be 0. Suppose for example that there were a nontrivial factor with $j = 0$. Since R is reduced, it follows that the commutator has a factor $u_\alpha(cx)$ in the product on the R.H.S. Putting $y = 0$ we derive a contradiction. $\square$

Lemma 7.2. *Suppose the Dynkin digram D of the root system R has connected components $D_1, \dots, D_r$. Then each D_i is the Dynkin diagram of a root system R_i and R is the disjoint union of the R_i , with every root in R_i orthogonal to every root in R_j for $i \neq j$. If D is connected, then R is irreducible in the sense that one cannot partition it into two nonempty subsets with every root in the first subset orthogonal to every one in the second.*

This should be assumed from the Lie algebra notes, The consequence of this last result for algebraic groups is

Theorem 7.2. *With notations as above, let G be semisimple and let $D_1, \dots, D_n$ be the irreducible components of the Dynkin diagram D of R , with D_i corresponding to the root system R_i and R the orthogonal disjoint union of the R_i . For each i there is a closed connected normal subgroup G_i of G with root system R_i . We have $[G_i, G_j] = 1$ for $i \neq j$. G is the product of G_i and the intersection of any G_i and the product of the others is finite. The groups G_i are also quasi-simple in the sense that they have no normal subgroups of positive dimension.*

Proof. For each i let T_i be the subtorus of T generated by the images of the coroots $\alpha^\vee$ for $\alpha \in R_i$. Then we take G_i to be the subgroup generated by T_i and U_α as α runs through the roots in R_i . For $\alpha \in R_k, \beta \in R_l$ with $k \neq l$, then $i\alpha + j\beta$ is not a root for $i, j > 0$. By proposition 7.5 we then have $[G_i, G_j] = 1$ for $i \neq j$. Hence the G_i are closed connected normal subgroups and G is their product. $\square$

7.1 Bruhat Decomposition

Now let Δ be a choice of simple roots. In the following we will study some basic results for general Coxeter groups

Proposition 7.6. *The simple reflections s_α for $\alpha \in \Delta$ generate W .*

For any $\beta \in R$, there is a product w of simple reflections such that $w\beta = \alpha \in \Delta$. Then one checks that $w^{-1}s_\alpha w = s_\beta$. Since the reflections s_β generate W by definition, so do the simple reflections. Hence given any $w \in W$ there is a unique minimum number h such that w is the product of h simple reflections. We denote h by $l(w)$, the **length of w** . If $w = s_1 \dots s_h$ where s_i are simple reflections and $h = l(w)$, then we say $s_1 \dots s_h$ is a reduced decomposition of w .

Given $w \in W$, set $R(w) = \{\alpha \in R^+ | w\alpha \in -R^+\}$

Lemma 7.3. *Let $w = s_1 \dots s_h$ be a reduced decomposition. Write α_i be the simple roots corresponding to s_i (remember s_i need not be distinct). Then $R(w) = \{\alpha_h, s_h\alpha_{h-1}, \dots, s_h s_{h-1} \dots s_2 \alpha_1\}$ so that in particular $R(w)$ has $h = l(w)$ elements.*

Proof. If $h = 1$, then for $\beta \in R^+$ $s_1\beta = \beta - \langle \beta, \alpha_1^\vee \rangle \alpha_1$ is still a positive root if $\beta \neq \alpha_1$, while $s_1\alpha_1 = -\alpha_1$. Hence this holds when $h = 1$. The same reasoning shows that $R(ws_\alpha) = s_\alpha R(w) \cup \{\alpha\}$ if $w\alpha \in R^+$, while $R(ws_\alpha) = s_\alpha(R(w) \setminus \{\alpha\})$ if $w\alpha \in -R^+$. The result follows at once by induction on h . $\square$

Lemma 7.4. *Still let $w = s_1 \dots s_h$ be a reduced decomposition of $w \in W$*

1. If α is simple, then $l(ws_\alpha) = l(w) - 1$ if $w\alpha \in -R^+$ and $l(ws_\alpha) = l(w) + 1$ if $w\alpha \in R^+$
2. Moreover, if $w\alpha \in -R^+$ there is a reduced decomposition of w ending with s_α
3. (the exchange condition) If $w = s'_1 \dots s'_h$ is another reduced decomposition, then there is i with $1 \leq i \leq h$ such that $s_1 \dots \hat{s}_i \dots s_h = s'_1 \dots \hat{s}'_i \dots s'_h$

Proof. (1) follows from the above formula of $R(ws_\alpha)$. For α given in (2), lemma 7.3 shows that $\alpha = s_h \dots s_{i+1} \alpha_i$ for some i and so $s_\alpha = s_h \dots s_{i+1} s_i s_{i+1} \dots s_h$. Hence $ws_\alpha = s_1 \dots \hat{s}_i \dots s_h$ which proves (2). (3) follows from (2) by letting $\alpha = \alpha'_h$ $\square$

Let s, t be simple reflections with $s \neq t$ and denote the order of st by $m(s, t)$. In root system we have $m(s, t) \in \{2, 3, 4, 6\}$. Also note that $m(s, t) = m(t, s)$. In the Lie algebra notes we have used the fact that the Weyl group can be presented as an abstract group by the generating set S of simple reflections with defining relations $s^2 = 1, (st)^{m(s, t)} = 1$. These last relations are called the **braid relations**. A consequence of the proof of this presentation of W is that given a map ϕ from S into a monoid M such that the $\phi(s)$ satisfy the braid relations, there is a unique extension of ϕ to W such that $\phi(w) = \phi(s_1) \dots \phi(s_h)$ for any reduced decomposition $s_1 \dots s_h$ of w

Note that there is a unique $w_0 \in W$ such that $w_0 R^+ = -R^+$. By lemma 7.4 (1) it is in fact the element of W with maximal length which equals $|R^+|$. We call w_0 **the long element of W** . Let $B_u = U$

Lemma 7.5. *For $w \in W$, the groups U_α with $\alpha \in R(w)$ generate a closed connected subgroup U_w of U normalized by T , and we have $U_w = \prod_{\alpha \in R(w)} U_\alpha$. Moreover, the product map $U_w \times U_{w_0 w} \rightarrow U$ is an isomorphism of varieties.*

Proof. U_w is connected closed and normalized by T since U_α are. Using proposition 7.5 we see that the product is a group and hence coincides with U_w . The second assertion then follows from proposition 7.4 $\square$

We give a uniform decomposition of any reductive group G using a Borel subgroup B and the Weyl group W relative to a torus T contained in B . Let $\{\dot{w}\}$ be a set of representatives in $N_G(T)$ of elements of W . Denote by $C(w)$ the double coset $B\dot{w}B$ (which is easily seen not depend on the choice of $\dot{w}$). This is an orbit of $B \times B$ acting on G , hence is open in its closure in G .

Lemma 7.6. *Let $w = s_1 \dots s_h$ be a reduced decomposition of $w \in W$. Let α_i be the simple root corresponding to s_i for each i . The morphism $\phi : \mathbb{A}^h \times B \rightarrow G$ with $\phi(x_1, \dots, x_h, b) = u_{\alpha_1}(x_1) \dot{s}_1 \dots u_{\alpha_h}(x_h) \dot{s}_h b$ defines an isomorphism $\mathbb{A}^h \times B \cong C(w)$. The map $(u, b) \mapsto u\dot{w}b$ is an isomorphism of varieties from $U_{w^{-1}} \times B$ to $C(w)$*

Proof. We have $C(w) = B\dot{w}B = U\dot{w}B$. Since elements of W conjugate one-parameter subgroups $(u_i(x))$ of G to one-parameter subgroups ([T.A10, exercise 8.1.12 (2)]) we have $\dot{w}^{-1}U_{w_0 w^{-1}}\dot{w} \subset B$, whence $C(w) = U_{w^{-1}}\dot{w}B$ by lemma 7.5. By lemma 7.3 we have $R(w^{-1}) = \{\alpha_1, s_1 \alpha_2, \dots, s_1 \dots s_{h-1} \alpha_h\}$. Then we have $U_{w^{-1}} = U_{\alpha_1}(\dot{s}_1 U_{w^{-1} s_1} \dot{s}_1^{-1})$ and $C(w) = U_{\alpha_1} \dot{s}_1 C(s_1 w)$. By induction on h we may assume that the assertion of the lemma holds for $s_1 w = s_2 \dots s_h$. It follows from the last formula that ϕ is surjective. Then ϕ is the composite of the isomorphism $\mathbb{A}^h \times B \rightarrow U_w \times B$ (lemma 7.5) and the morphism $(u, b) \mapsto u\dot{w}b$. We have to prove that the latter morphism is an isomorphism or, equivalently, that $(u, b) \mapsto \dot{w}^{-1}u\dot{w}b$ defines an isomorphism of $U_{w^{-1}} \times B$ onto $\dot{w}C(w)$. This can be done by viewing both spaces as homogeneous spaces for $U_{w^{-1}} \times B$ and theorem 4.3 (3) $\square$

Lemma 7.7. *Let $w \in W, s \in S$ where S is the set of simple reflections. Then $C(s)C(w) = C(sw)$ if $l(sw) = l(w) + 1$ and $C(s)C(w) = C(w) \cup C(sw)$ if $l(sw) = l(w) - 1$.*

Proof. Let α be the simple root corresponding to s . Then lemma 7.6 gives $C(s) = U_\alpha \dot{s} B$, whence $C(s)C(w) = U_\alpha \dot{s} C(w)$. If $l(sw) = l(w) + 1$ then this follows from lemma 7.6. If $l(sw) = l(w) - 1$ then we have $C(s)C(w) = C(s)C(s)C(sw)$. The lemma will follow if we show that $C(s)C(s) = C(e) \cup C(s)$. Using lemma 6.3 we see that $C(e) \cup C(s)$ is the group G_α defined in the previous section, and by theorem 6.2 the quotient of this group by its radical is SL_2 or PSL_2 . Then we can check this lemma for $G = SL_2$ directly. $\square$

We finally give the decomposition we are after:

Theorem 7.3. *(Bruhat's lemma) G is the disjoint union of the double cosets $C(w)$ for $w \in W$.*

Proof. Set $G_1 = \cup_{w \in W} C(w)$. By lemma 7.7 we have $C(s)G_1 = G_1$ for $s \in S$. It follows that G_1 is stable under the multiplication by the elements of T , U_α , $U_{-\alpha}$ for simple roots α (note that $T \subset B \subset G_1$). Hence $G = G_1$, since these groups generate G .

Suppose $C(w) \cap C(w') \neq \emptyset$ for some $w, w' \in W$. Then $C(w) = C(w')$. By lemma 7.6 we have $\dim C(w) = l(w) = \dim B$, and hence $l(w) = l(w')$. We may assume $l(w) > 0$. Then there is $s \in S$ such that $l(sw) = l(w) - 1$. By lemma 7.7, $C(sw) \subset C(s)C(w) = C(s)C(w') \subset C(w') \cup C(sw')$. Hence $C(sw) = C(w')$ or $C(sw) = C(sw')$ since these double cosets are irreducible. Since $l(sw) \neq l(w')$, we must have $C(sw) = C(sw')$. By induction on $l(w)$ we have $sw = sw' \implies w = w'$ since $l(sw) = l(w) - 1$. $\square$

Consequently, elements in G can be uniquely written in the form uwb for some $w \in W, u \in U_{w^{-1}}$ and $b \in B$.

Corollary 7.4. *The intersection of any two Borel subgroups of G contains a maximal torus.*

Proof. Assume the two Borel subgroups are $B, B' = gBg^{-1}$. Write $g = bwb'$ for some $w \in W, b, b' \in B$. Then $bTb^{-1} \subset B \cap B'$. $\square$

Note that since Borel subgroups containing a fixed maximal torus T are in bijection to Weyl group elements w , we can express the relation between any two Borel subgroups (sometimes called the **attitude** of one with respect to the other) by such an element.

Corollary 7.5. *There is a unique open double coset $C(w_0)$ in G with w_0 the longest element of W .*

In fact, $C(w_0)$ is the unique double coset with dimension $\dim G$ by lemma 7.5, lemma 7.6 and corollary 7.3 (2). It is then open in its closure G by lemma 1.1. It is often called the **big cell**.

7.2 Flag Variety

Recall the variety $\mathfrak{B}$ of a fixed Borel subgroup is isomorphic to the homogeneous space G/B . In the setting of G be reductive, we call it **the flag variety**. To understand its name, recall that a flag of a finite-dimensional vector space V over k is a chain of subspaces $V_0, V_1 \dots$ such that each V_i is properly contained in V_{i+1} . The flag is **complete** if this chain is not properly included in a larger one, i.e. $V_0 = 0$, $\dim V_i = i$, and the chain ends with $V_n = V$. The Lie-Kolchin Theorem shows at once that an arbitrary Borel subgroup B of $G = GL_n$ is exactly the stabilizer of a complete flag $F = V_0 \subset \dots \subset V_n = k^n$, in the sense that $x \in G$ lies in B if and only if $xV_i = V_i$ for all i . Moreover, the subgroup B determines and is determined by the flag F . Similarly, if instead F is only a partial (possibly incomplete) flag,

then its stabilizer in G is a parabolic subgroup. It turns out that all parabolic subgroups arise in this way.

The Bruhat's decomposition implies a similar result for the flag varieties. Let $\pi : G \rightarrow \mathfrak{B}$ be the canonical map. Put $X(w) = \pi C(w)$

Proposition 7.7. *1. $\mathfrak{B}$ is the disjoint union of the locally closed subvarieties $X(w)$ for $w \in W$. They are the B -orbits in B . In particular there are only finitely many such orbits.*

2. $X(w)$ is an affine variety isomorphic to $\mathbb{A}^{l(w)}$

3. $X(w)$ contains a unique point x_w fixed by T

4. There is a cocharacter λ of T such that for all $x \in X(w)$ we have $\lim_{a \rightarrow 0} \lambda(a)x = x_w$. That is, the morphism λ admits a unique extension to k which maps 0 to x_w .

Proof. (1) follows directly from the Bruhat's decomposition. (2) follows from lemma 7.6. The fixed point in (3) is just $\pi(w)$. For λ , choose any λ with $\langle \lambda, \alpha \rangle > 0$ for $\alpha \in R^+$. Then $\lambda(a)u_\alpha(b)\lambda(a)^{-1} = u_\alpha(a^{\langle \lambda, \alpha \rangle}b)$ for $a \in k^*, b \in k$. From this we see that for $u \in U_{w^{-1}}$, $\lim_{a \rightarrow 0} \lambda(a)u\lambda(a)^{-1} = e$, which proves (4) $\square$

This result gives a stratification of $\mathfrak{B}$, i.e. a decomposition of $\mathfrak{B}$ into locally closed subsets called **strata**. The strata $X(w)$ are affine spaces, called **Schubert cells** (or Bruhat cells, as in the text). The closure $S(w) = \overline{X(w)}$ is called a **Schubert variety**. The open stratum $X(w_0)$ is once again called the big cell; its translates $gX(w_0)$ cover $\mathfrak{B}$.

Lemma 7.8. *π has local sections*

This follows from last proposition and lemma 7.6 by identifying $U_{w^{-1}}$ with $X(w)$

Return to G . Note that the closure $\overline{C(w)}$ of $C(w)$ is the union of various B -orbits $C(x)$. We define a partial order on W via $x \leq w$ if $C(x) \subset \overline{C(w)}$. It is immediate that this really is a partial order (called the **Bruhat** or **Bruhat-Chevalley order**) and that we have $x \leq w$ if and only if the Schubert variety $S(x)$ is contained in $S(w)$. *There is a beautiful combinatorial description of this order in terms of reduced decompositions in W .*

Fix a such decomposition $s_1 \dots s_h$ of $w \in W$. Denote by I_w the set of $x \in W$ that can be written in the form $s_{i_1} \dots s_{i_m}$ for some indices $1 \leq i_1 \leq \dots \leq i_m \leq h$. We will see that this definition is independent of the choice of reduced decomposition of w .

Proposition 7.8. *For $x, w \in W$, $x \leq w \iff x \in I_w$*

Proof. We first need a claim

Claim: Let P and Q be parabolic subgroups of G with $P \subset Q$. Let X be a closed subset of G such that $XP = X$ (i.e. X is P -stable). Then XQ is closed. ([T.A10, exercise 6.2.11 (5)])

Let π be the canonical quotient map. Then $\pi(X)$ is closed and hence complete in G/P since $\pi^{-1}\pi(X) = X$ by the P -stability of X . Hence $\pi(XQ)$ is also closed and complete in G/P , and its preimage XQ is closed in G

Now let $P_i = C(e) \cup C(s_i)$. [T.A10, proposition 8.4.3] shows that these P_i are parabolic subgroups. By the claim we see that $P_1 \dots P_h$ is an irreducible closed subgroup of G . By lemma 7.7 it is in fact the union of $C(y)$ for $y \in I_w$. Then $C(w)$ is the unique one with maximal dimension among these double cosets lemma 7.5 and lemma 7.6. Then $\overline{C(w)} \subset P_1 \dots P_h$. Since both sets are irreducible, closed, and have the same dimension, they coincide. In particular, the definition of I_w is indeed independent of the choice of $s_1 \dots s_h$. $\square$

8 The Isomorphism Theorem

In this chapter G still denotes a connected reductive linear algebraic group over k and T a maximal torus. We will show that the root datum of G determines the group up to isomorphism, and that every abstract root datum is the root datum of a reductive group.

First we fix a realization of $\{u_\alpha\}$ of the root system R in G . Given $\alpha, \beta \in R$ recall that by proposition 7.5 we have the structure constants $c_{\alpha,\beta,i,j}$ satisfying $[u_\alpha(x), u_\beta(y)] = \prod_{i\alpha+j\beta \in R} u_{i\alpha+j\beta}(x^i y^j)$ for $x, y \in k$. If $\{c'_{\alpha,\beta,i,j}\}$ is another set of structure constants arising from a different realization $\{u'_\alpha\}$ of R , by proposition 7.3 (4) we deduce that there are $c_\alpha \in k$ for $\alpha \in R$ with $c_\alpha c_{-\alpha} = 1$ such that $c'_{\alpha,\beta,i,j} = c_\alpha^{-1} c_\beta^{-1} c_{i\alpha+j\beta} c_{\alpha,\beta,i,j}$. In this case we call these two structure constants are equivalent. . For convenience we set $c_{\alpha,\beta,1,0} = c_{\alpha,\beta,0,1} = 1$ for all roots α, β

Now we lay put several lemmas without proofs to show that a root datum $D = (X, R, X^\vee, R^\vee)$ determines a reductive group up to isomorphism. Recall that we set $n_\alpha = u_\alpha(1)u_{-\alpha}(-1)u_\alpha(1)$ which normalizes T and represents the reflection s_α . For another $\beta \in R$, $x \in k$, define $u'_\beta(x) = n_\alpha u_{s_\alpha \beta}(x) n_\alpha^{-1}$. Then $\{u'_\beta\}$ is another realization of R , and hence there is $d_{\alpha,\beta} \in k^*$ such that $n_\alpha u_\beta(x) n_\alpha^{-1} = u_{s_\alpha \beta}(d_{\alpha,\beta} x)$. If $\beta \neq \pm\alpha$, consider the α -string through β in R $\beta - c\alpha, \beta - c\alpha + \alpha, \dots, \beta + b\alpha$ for some $c, b \in \mathbb{N}$

Lemma 8.1. 1. $d_{\alpha,\beta} = \sum_{i=0, c-b}^c (-1)^i c_{-\alpha,\beta,i,1} c_{\alpha,\beta-i\alpha,i+b-c,1}$

$$2. d_{-\alpha,\beta} = (-1)^{\langle \beta, \alpha^\vee \rangle} d_{\alpha,\beta}, d_{\alpha,\beta} d_{-\alpha,-\beta} = d_{\alpha,\beta} d_{\alpha,s_\alpha \beta} = (-1)^{\langle \beta, \alpha^\vee \rangle}, d_{\alpha,\alpha} = -1, d_{\alpha,\beta} d_{\alpha,-\beta} = 1$$

$$3. n_\alpha n_\beta n_\alpha^{-1} = (s_\alpha \beta)^\vee (d_{\alpha,\beta}) n_{s_\alpha \beta}$$

Proof. See [T.A10, lemma 9.2.2] □

Using this lemma, we have the following result on the realization of an irreducible R of rank 2 with simple roots α, β

Proposition 8.1. 1. We can normalize the realization $\{u_\gamma\}$ such that all structure constants lie in $\mathbb{Z}$ (or in $\mathbb{Z}/p$ in characteristic p). We have $c_{\alpha,\beta,i,j} = 1$ for $i, j > 0, i\alpha + j\beta \in R$

2. All $d_{\alpha,\gamma}, d_{\beta,\gamma}$ are ± 1 for positive roots γ

3. If R has type G_2 then $c_{\beta,3\alpha+\beta,1,1} = 1$ The above properties uniquely determine all structure constants for rank-2 systems. Moreover we have $c_{\alpha,\beta,i,j} = c_{w\alpha,w\beta,i,j}$ for all $w \in W$

Proof. See [T.A10, proposition 9.2.5] □

Now we explore properties of the element n_α defined above. Set $t_\alpha = \alpha^\vee(-1)$. Then $n_\alpha^2 = t_\alpha$. Still denote by $m(\alpha, \beta)$ the order of $s_\alpha s_\beta$

Proposition 8.2. Let α, β be simple roots. Then braid relations $n_\alpha n_\beta n_\alpha \dots = n_\beta n_\alpha n_\beta \dots$ hold. The number of factors on both sides are $m(\alpha, \beta)$

Proof. This is reduced to rank-2 root system by considering the subsystem spanned by α, β . See [T.A10, proposition 9.3.2] □

Now we are ready to give a presentation of (G, T) as an abstract group given its root datum $(X, R, X^\vee, R^\vee)$, a realization $\{u_\alpha\}$ and structure constants $c_{\alpha,\beta,i,j}$. Fix a selection of simple roots Δ in R . Then we may assume the realization $\{u_\alpha\}$ is normalized as in proposition 8.1 and are then uniquely determined up to equivalence. It then turns out that $c_{\alpha,\beta,i,j} = \pm(c+1)$, where c is the largest nonnegative integer such that $\alpha - c\beta \in R$. ([T.A10, proposition 9.5.3]). We begin by realizing a maximal torus $\mathbb{T} = \text{Hom}(X, k^*)$. For $\chi \in X$, define a homomorphism $\bar{\chi} : \mathbb{T} \rightarrow k^*$ via $\bar{\chi}(t) = t(\chi)$.

For $\lambda \in \chi^\vee$, define a homomorphism $\bar{\lambda} : k^* \rightarrow \mathbb{T}$ via $\bar{\lambda}(x)(\chi) = x^{\langle \chi, \lambda \rangle}$ for $x \in k^*$. The Weyl group W acts on $\mathbb{T}$ via $w \cdot t(\chi) = t(w^{-1}(\chi))$

Now for $\alpha \in R, x \in k$, we have a generator $u_\alpha(x)$. For $x, y \in k$ we impose the relations $u_\alpha(x)u_\alpha(y) = u_\alpha(x+y)$ and $[u_\alpha(x), u_\beta(y)] = \prod_{i\alpha+j\beta \in R, i,j>0} u_{i\alpha+j\beta}(c_{\alpha,\beta,i,j}x^i y^j)$ for $\alpha \neq \pm\beta$ using the given structure constants. It turns out to be enough to impose these relations only on roots α, β both lying in the span of two simple roots in Δ . Then we have relations $tu_\alpha(x)t^{-1} = u_\alpha(\bar{\alpha}(t)x)$ for $t \in \mathbb{T}, x \in k, \alpha \in R$. Set $n_\alpha = u_\alpha(1)u_{-\alpha}(-1)u_\alpha(1)$. Then we require $n_\alpha^2 = t_\alpha$ where $t_\alpha(\chi) = (-1)^{\langle \chi, \alpha^\vee \rangle}$ for $\chi \in X, n_\alpha u_\alpha(x)n_\alpha^{-1} = u_{-\alpha}(-x)$. Further we require $u_\alpha(x)u_{-\alpha}(-x^{-1})u_\alpha(x) = \bar{\alpha}^\vee(x)n_\alpha$ for $x \in k$, and finally $n_\alpha n_\beta n_\alpha \dots = n_\beta n_\alpha n_\beta$ with $m(\alpha, \beta)$ factors on each side.

Let $\mathbb{G}$ be the group generated by $\mathbb{T}$ and $u_\alpha(x)$ for $\alpha \in R, x \in k$, subject to the above relations

Theorem 8.1. *The isomorphism $T \xrightarrow{\sim} \mathbb{T}$ extends to an isomorphism $G \cong \mathbb{G}$ sending $u_\alpha(x) \mapsto u_\alpha(x)$ for $\alpha \in R, x \in k$*

Proof. See [T.A10, theorem 9.4.3] □

In particular, any two reductive groups G, G' with isomorphic root data D, D' are isomorphic. Note however that this argument does not amount to a proof that a reductive group exists with any given root datum D , as it is not a priori clear that the abstract group G defined above has the structure of an affine variety

*We do however get some extra mileage out of this argument. First of all, since the structure constants $c_{\alpha,\beta,i,j}$ lie in $\mathbb{Z}$ (or in some $\mathbb{Z}/p$) we see that an analogue of G exists over any basefield k , not necessarily algebraically closed. We will say more later about the groups that arise in this way. They are called **Chevalley groups**.*

*Next, the proof shows that any automorphism ϕ of the torus T preserving R induces an automorphism of the corresponding algebraic group G , unique up to conjugation by some $t \in T$. This is obvious for automorphisms of T coming from the action of W , since any such automorphism is realized by conjugation by a suitable element of the normalizer $N_G(T)$. But now some automorphisms of R do not arise in this way, namely those arising from nontrivial automorphisms of the Dynkin diagram D' corresponding to R . We will say more about such automorphisms, called **diagram automorphisms**, later. They play a crucial role in the proof that given any root datum D there is an algebraic group with that datum*

9 The Existence Theorem

In this chapter we will show that there exists a connected reductive linear algebraic group G over k with any given root datum $(X, R, X^\vee, R^\vee)$. First we notice that it can be reduced to the case that R spans X , so that the groups G we construct will be semisimple and of adjoint type.

First we define the Lie algebra of G . Set $\mathfrak{t} = k \otimes X^\vee$ and $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} k e_\alpha$, where e_α are linearly independent generators. Define the Lie bracket via

$$[u, u'] = 0, [u, e_\alpha] = \langle \alpha, u \rangle e_\alpha, [e_\alpha, e_\beta] = c_{\alpha,\beta} e_{\alpha+\beta}, [e_\alpha, e_{-\alpha}] = 1 \otimes \alpha^\vee$$

for $u, u' \in T, \alpha \neq \pm\beta$, and $e_{\alpha+\beta} = 0$ for $\alpha + \beta \notin R$. Here $c_{\alpha,\beta}$ are some structure constants to be specified

Now we only consider **simply laced root system**, i.e. roots are all of the same length. Recall that this holds if and only if there are no multiple bonds in the Dynkin diagram of R , and we normally take $(\alpha, \alpha) = 2$ for all $\alpha \in R$. For a simply laced R , it is easy to see that all root strings have length ≤ 1 . From this we can deduce that for $\alpha, \beta \in R$ and $\epsilon = \pm 1$, $\alpha + \epsilon\beta \in R \iff \langle \alpha, \beta^\vee \rangle = -\epsilon$. Note that there

is a $\mathbb{Z}$ -valued bilinear function f on X satisfying $(x, y) \equiv f(x, y) + f(y, x) \pmod{2}$, $f(x, x) \equiv \frac{1}{2}(x, x) \pmod{2}$.

Fix a choice of positive roots R^+ and put $\epsilon(\alpha) = \pm 1$ for $\alpha \in \pm R^+$. Finally we set $c_{\alpha, \beta} = \epsilon(\alpha)\epsilon(\beta)(-1)^{f(\alpha, \beta)}$ if $\alpha, \beta, \alpha + \beta \in R$ and 0 else. Using the properties of f , we have the following results

Lemma 9.1. 1. $c_{\alpha, \beta} = -c_{\beta, \alpha}$, and $c_{-\alpha, \beta}c_{\alpha, -\alpha+\beta} + c_{\beta, \alpha}c_{-\alpha, \alpha+\beta} = \langle \beta, \alpha^\vee \rangle$

2. If $\alpha, \beta, \gamma \in R$ linearly independent then $c_{\alpha, \beta}c_{\alpha+\beta, \gamma} + c_{\beta, \gamma}c_{\beta+\gamma, \alpha} + c_{\gamma, \alpha}c_{\gamma+\alpha, \beta} = 0$

As a consequence the Jacobi identity holds on basis vectors of $\mathfrak{g}$, so by linearity holds in general. Thus $\mathfrak{g}$ is indeed a Lie algebra.

Recall that a derivation on a Lie algebra $\mathfrak{h}$ is a linear map D on $\mathfrak{h}$ satisfying $D[x, y] = [Dx, y] + [x, Dy]$. In particular, $ad_x : y \mapsto [x, y]$ is a derivation on $\mathfrak{h}$. Such derivations are called **inner**. In the Lie algebra notes we have shown that any derivation on the Lie algebra is inner.

Now let T be a torus with character group X . Let T act on $\mathfrak{g}$ via $t \cdot u = u$ for $u \in \mathfrak{t}$, $t \cdot e_\alpha = \alpha(t)e_\alpha$ for $\alpha \in R$. This defines an isomorphism of T onto a closed subgroup of $GL(\mathfrak{g})$. Put $X_\alpha = ad_{e_\alpha}$. Note that $X_\alpha^3 = 0$ since the root string has length ≤ 1 . Denote by $X_\alpha^{(2)}$ the linear map sending all $u \in T$ and e_β for $\beta \neq -\alpha$ to 0 and $e_{-\alpha}$ to $-e_\alpha$. Then $2X_\alpha^{(2)} = X_\alpha^2$ since $\alpha^\vee(\alpha) = 2$. Define linear maps $u_\alpha(x)$ of $\mathfrak{g}$ by letting $u_\alpha(x) = 1 + xX_\alpha + x^2X_\alpha^{(2)}$ for $x \in k$. It follows that $u_\alpha(x+y) = u_\alpha(x)u_\alpha(y)$, whence an isomorphism u_α of $\mathbb{G}_a$ onto a closed subgroup U_α of $GL(\mathfrak{g})$. We denote by G the subgroup of $GL(\mathfrak{g})$ generated by T and the U_α . Then G is closed and connected.

Proposition 9.1. G is reductive and T is a maximal torus of G . The root system of (G, T) is R .

Proof. Indeed, G acts on $\mathfrak{g}$ by automorphisms by the construction. As the differential of an automorphism is a derivation which is inner, we deduce that $L(G)$ is contained in $\mathfrak{g}$. Since the Lie algebra of U_α is kX_α , $L(G)$ contains $L(T) \oplus \bigoplus_{\alpha \in R} kX_\alpha$, and hence $L(G) = \mathfrak{g}$. The group G_α defined in chapter 4 contains $U_\alpha, U_{-\alpha}$ and stabilizes the subspace generated by $1 \otimes \alpha^\vee, e_\alpha, e_{-\alpha}$. And the group generated by the restrictions of $U_{\pm\alpha}$ to that space is not solvable. Hence α is a root of (G, T) . Note that $\dim R_u(G) = \dim G - \dim T - |R| = 0$, so G is reductive. $\square$

We conclude by remarking that another construction of the simply connected group G having a given root system R arises as the universal covering of G as an abstract topological group, bearing in mind that any such covering always has a topological group structure and that its deck transformations all arise from elements of its center (which turns out for algebraic groups to be isomorphic to the quotient of the weight lattice of R by its root lattice, as previously noted). The other groups having root system R , including the adjoint group, are then all quotients of this one by a finite central subgroup. We also observe that the non-semisimple adjoint group G' corresponding to a datum $(X, R, X^\vee, R^\vee)$ for which X is not spanned by R but is instead the direct product of the abelian subgroup Q spanned by R and its orthogonal under the form $(,)$ is easily constructed from the group G with datum $(Q, R, Q^\vee, R^\vee)$ by taking the direct product of this group and a suitable torus.

Now we construct the adjoint groups of non-simply laced types. First note that the Dynkin diagrams of the simply laced type A_n, D_n and E_6 all admit a nontrivial automorphism, i.e. a bijection of the set of vertices such that two nodes are adjacent iff their images are. For example, for A_n we can just reverse the string of vertices; for D_n , we can interchange the two non-adjacent vertices at one end of the diagram while fixing all the others, while for D_4 we can also permute the three outer vertices arbitrarily while fixing the inner one. In all cases note that if we 'fold' these Dynkin diagrams by compressing the s -orbits to vertices and combining the edges from moved simple roots to the unmoved ones to make

double or triple edges, we essentially get Dynkin diagrams of type C_n, B_n, F_4 or G_2 from the simply laced ones. For this reason this operation is called **folding**.

Let R, R^+ as above with Dynkin diagram D and the automorphism s . Note that s uniquely extends to an automorphism of Q , the $\mathbb{Z}$ -span of R in V , preserving R, R^+, D . We also have the bilinear form f on Q defined above arising from the basis of Q of simple roots in D , ordered so that simple roots in the same s -orbit occur consecutively in the order. In this way we ensure that f is s -invariant. Note that s has order 2 or 3, and that if it has order 3 there is just one s -orbit of simple roots of cardinality 3. A simple case-by-case verification shows that in all cases we have

Lemma 9.2. 1. $(1-s)Q \cap R = \emptyset$

2. If $\alpha, \beta \in R$ and $\alpha - \beta \in (1-s)X$ then α, β lies in the same s -orbit

3. A root is orthogonal to the other roots in its orbit.

Also s defines an automorphism σ on the Lie algebra $\mathfrak{g}$ defined above via $\sigma(1 \otimes \alpha^\vee) = 1 \otimes s \cdot \alpha^\vee, \sigma(e_\alpha) = e_{s\alpha}$ since f is s -invariant. It further induces an automorphism of the adjoint group G of Q, R stabilizing the maximal torus T , which we still denote by σ . Then σ has the same order as s and we have $\sigma(u_\alpha(x)) = u_{s\alpha}(x)$ for $\alpha \in R, s \in k$

Set $Q^s = Q/(1-s)Q$. Since s permutes the basis of Q , it follows that Q^s is torsion-free, and so Q^s is a free abelian group. Take $\mathcal{O}$ a s -orbit in R and let $\alpha_{\mathcal{O}}$ be the coset of α in Q^s for any $\alpha \in \mathcal{O}$. Clearly $\alpha_{\mathcal{O}}$ is independent of the choice of $\alpha \in \mathcal{O}$. Set $\alpha_{\mathcal{O}}^\vee = \sum_{\alpha \in \mathcal{O}} \alpha^\vee$ and $Q^{s\vee}$ the submodule of $Q^\vee$ annihilating $(1-s)Q$. Then $\alpha_{\mathcal{O}}^\vee \in Q^{s\vee}$. Let $R^s, R^{s\vee}$ the set of $\alpha_{\mathcal{O}}$ and $\alpha_{\mathcal{O}}^\vee$ respectively with $\mathcal{O}$ running through the s -orbits. The one checks that $(Q^s, R^s, Q^{s\vee}, R^{s\vee})$ is a root datum ([T.A10, lemma 10.3.4]). We say it is obtained from $(Q, R, Q^\vee, R^\vee)$ by folding according to s .

Denote fixed point sets under an automorphism by a superscript. Then the existence theorem for non-simply-laced root systems follows from

Proposition 9.2. $(G^\sigma)^0$ is a connected, reductive group with maximal torus T^σ . The root datum of $(G^\sigma)^0, T^\sigma$ is isomorphic to $(Q^s, R^s, Q^{s\vee}, R^{s\vee})$

Proof. Set $H = (G^\sigma)^0$. Note that the character group of T^σ is precisely Q^s which is a free abelian group. Hence T^σ is a torus. By lemma 9.2 there is no root acting trivially on T^σ . Then we see that the centralizer of T^σ in G is T , which implies T^σ is a maximal torus of H

For $\alpha_{\mathcal{O}}^s \in R^s$ put $u_{\alpha_{\mathcal{O}}}(x) = \prod_{\beta \in \mathcal{O}} u_\beta(x)$ for $x \in k$ where $\{u_\beta\}$ is the realization arising in the construction of G . Then $u_{\alpha_{\mathcal{O}}}$ defines an isomorphism from $\mathbb{G}_a$ onto some closed subgroup of H . An argument similar to the proof of proposition 9.1 shows that $\alpha_{\mathcal{O}}$ are roots of H, T^σ , whence $\dim H \geq \dim T^\sigma + |R^s|$. On the other hand, the Lie algebra $\mathfrak{h} = L(H)$ is a subalgebra of the fixed-point algebra $\mathfrak{g}^\sigma$ whose dimension is $\dim T^\sigma + |R^s|$. Since both $\dim T^\sigma$ and $\dim \mathfrak{h}$ equal the number of s -orbits in the Dynkin diagram, it follows that H is reductive □

Since we have shown that for a non-simply laced system it can be obtained from a simply laced one by folding according to some automorphism s , we have proven that there is connected reductive group with any given datum.

10 Chevalley Groups and Real Lie Groups

(This chapter completely follows the lecture notes at UW by Professor Monty McGovern)

We generalize the construction of adjoint groups from root systems beyond the simply laced case. The construction works over an arbitrary basefield k and gives rise to the Chevalley groups. We also show how to construct certain real Lie (but non-algebraic) groups and Lie algebras from such systems, starting from the complex ones already constructed.

Given a root system R , our first task is to construct a Lie algebra with this root system. We have seen how to do this for simply laced system. In general, still set $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} k e_\alpha$ with Lie bracket as defined in previous chapter with some structure constants $c_{\alpha,\beta}$ to be specified. A well-known result of Chevalley asserts that this can always be done and we can arrange that $c_{\alpha,\beta} = -c_{-\alpha,-\beta}$. Whenever this last property holds we automatically have $c_{\alpha,\beta} = \pm(r+1)$, where r is the largest nonnegative integer such that $\beta - r\alpha \in R$. *This follows from a more elaborate version of the construction given above in the simply laced case. Or else we can give an abstract presentation of $\mathfrak{g}$ by generators and relations (following Serre) and then use an automorphism of $\mathfrak{g}$ of order 2 to find root vectors e_α with the desired properties.*

The formula for $c_{\alpha,\beta}$ up to sign shows that the coefficients of all powers of x in the series for $\exp ad_{x e_\alpha}(e_\beta)$ are binomial coefficients up to sign for any $x \in k, \alpha, \beta \in R$. Since $ad_{x e_\alpha}$ is nilpotent, the series for $\exp ad_{x e_\alpha}$ is well-defined on $\mathfrak{g}$ for any $x \in k$ regardless the characteristic. Then one can form the group of automorphisms of $\mathfrak{g}$ generated by the torus T with character group $\mathbb{Z}R$ together with groups U_α defined as before. This is called the **Chevalley group** attached to R and k . Chevalley showed that if R is irreducible then this group is such that every proper normal subgroup is central, apart from a few exceptions for very small k . *By proving this, he was able to exhibit several new families of previously unknown finite simple groups. In fact, the alternating groups, the finite Chevalley groups, and modifications of these groups called twisted Chevalley groups turn out to account for all finite simple groups with just 26 exceptions (the so-called sporadic simple groups).*

The Lie algebra $\mathfrak{g}$ constructed as above is called split. If R is irreducible and the characteristic of k is either 0 or large enough, then $\mathfrak{g}$ is simple, i.e. has no nonzero proper ideal

We conclude with a construction of a very different class of Lie algebras and adjoint groups. Given the root system R , let Δ be a choice of simple roots. Let $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathbb{C} e_\alpha$ as above with base field $\mathbb{C}$. Fix a square root i of -1 and let $\mathfrak{k} = \mathfrak{g}\mathbb{R}$ be the real span of $i\alpha^\vee, e_\beta - e_{-\beta}, i(e_\beta + e_{-\beta})$ as α, β run over Δ . Since the structure constants $c_{\gamma,\delta}$ are real with $c_{\gamma,\delta} = -c_{-\gamma,-\delta}$, one checks immediately that $\mathfrak{k}$ is closed under the Lie bracket. There is however no analogue of root space decomposition for $\mathfrak{k}$. In fact it turns out for any $x \in k$ that ad_x acts on the complexification $\mathfrak{k} \otimes \mathbb{C}$ of $\mathfrak{k}$ with purely imaginary eigenvalues, so that even though ad_x acts semisimply on $\mathfrak{k}$, it does not act diagonally unless $x = 0$.

Given $\mathfrak{k}$ we can define its adoint group K to be generated by $\exp ad_{xi\alpha^\vee}, \exp ad_{x(e_\beta - e_{-\beta})}, \exp ad_{xi(e_\beta + e_{-\beta})}$ for $x \in k, \alpha, \beta \in \Delta$. However, this time we don't have ad_x to be nilpotent for $x \in \mathfrak{k}$. So we need the infinite series definition of $\exp ad_x$ and the completeness of the real field to define K .

To see this group is compact, recall that the **Killing form** κ on a Lie algebra $\mathfrak{h}$ is defined by $\kappa(x, y) = \text{tr}(ad_x ad_y)$. We have seen that it is indeed a symmetric invariant bilinear form, and it is nondegenerate if and only if $\mathfrak{h}$ is semisimple. Now give $\mathfrak{g}$ as above, we know there is a complex algebraic group G with maximal torus T and root datum $(X, R, X^\vee, R^\vee)$ relative to T whose Lie algebra is $\mathfrak{g}$. The complexification $\mathbb{C}X$ of the character group X of T then identifies with $\mathfrak{t}^*$. Then the biliner form $(,)$ previously defined on $\mathbb{C}X$ induces a form on $\mathfrak{t}^*$, which then can be transferred to $\mathfrak{t}$ by identifying $\alpha^\vee$ with $2\alpha/(\alpha, \alpha)$. This transferred form turns out to coincide with the restriction of the Killing form to $\mathfrak{t}$. We also have $\kappa(e_\alpha, e_{-\alpha}) = 2/(\alpha, \alpha)$ and it is easy to check from the definition that $\mathbb{C}e_\alpha$ and $\mathbb{C}e_\beta$ are orthogonal under this form whenever $\beta \neq \pm\alpha$.

A result shows that the Killing form is negative definite when restricted to $\mathfrak{k}$. As the action of K on $\mathfrak{k}$ (or on $\mathfrak{g}$) clearly preserves this form, we can realize K as a closed subgroup of the orthogonal group $O_n(\mathbb{R})$, where n is the dimension of $\mathfrak{k}$. Hence K is indeed compact as a topological group. This property compensates somewhat for the lack of a root space decomposition of $\mathfrak{k}$. The group K arising

in this way from $G = SL_n(\mathbb{C})$ is the special unitary group SU_n . If $G = SO_n(\mathbb{C})$ then K is the real orthogonal group $SO(n, \mathbb{R})$

There are also the split real Lie groups, which are exactly the Chevalley groups attached as above to a root system R and the real field. These are the special linear groups $SL_n(\mathbb{R})$ in type A and symplectic groups $Sp_{2n}(\mathbb{R})$ in type C . In types B and D , one more generally has the (disconnected) groups $SO_{p,q}$ and their identity components $SO_{p,q}^0$ for any positive integers p and q , defined to be the symmetry groups of nondegenerate symmetric forms $(\cdot, \cdot)_{p,q}$ of signature (p, q) on $\mathbb{R}^{p+q}$. These last groups are split if and only if either $p = q$ or $p = q \pm 1$.

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